PEDIATRIC MOONSHOT

# Pediatric Cancer Atlas
# The Complete Pediatric Cancer Challenge

104 Challenges

AI can accelerate diagnosis and treatment for children rurally, locally and globally

“If you know the enemy and know yourself, you need not fear the result of a hundred battles.”
— Sun Tzu, The Art of War

To defeat childhood cancer, we must first define it. If we are going to make real progress against childhood cancer, we have to start with a clear understanding of what we are actually trying to solve. The Pediatric Cancer Atlas brings together 104 distinct cancers that affect children worldwide, spanning leukemias, lymphomas, brain and central nervous system tumors, solid tumors, and a number of rare malignancies. For each, it outlines the underlying biology, how frequently it occurs, how it is diagnosed, and how it is currently treated. It also distinguishes between what is established as standard of care and what is still emerging, which is important because much of the field is actively evolving.

Taken together, this is a picture of both progress and persistent challenge. Over the past several decades, advances in cooperative group trials, risk stratification, and multimodal therapy have dramatically improved outcomes for many children with cancer. In high income countries, cure rates now exceed 80 percent for several major diseases. At the same time, there are cancers where progress has been far more limited, and even for those who are cured, the long term consequences of treatment can shape the rest of their lives.

What becomes clear very quickly is that the barriers are not only biological. They are structural. Most pediatric cancers are rare, which makes it difficult to generate the volume of data needed to drive discovery and to run efficient clinical trials. Expertise is concentrated in specialized centers, which creates real access challenges for patients and families who are not near those institutions. Data is often fragmented across systems, which limits our ability to learn from each patient in a meaningful way. And these challenges are even more pronounced in low and middle income settings, where delays in diagnosis and limited access to treatment continue to impact outcomes.

Within this context, there is growing interest in how advances in data science and artificial intelligence might help address some of these gaps. In practice, this is already happening across the research community. Investigators are using computational approaches to better interpret imaging, to integrate genomic and clinical data, and to identify patterns that would otherwise be difficult to see, particularly in rare diseases. There is also increasing focus on how these tools might support more efficient clinical trial enrollment, which remains one of the major barriers to moving promising discoveries into patient care.

At the same time, it is important to be clear about what these approaches can and cannot do. They are not a replacement for the underlying science or for the clinical infrastructure required to care for patients. Their value will depend on how thoughtfully they are developed and how well they are integrated into the broader research and care ecosystem.

The Atlas is intended to serve as a foundation for that work. By laying out the landscape of pediatric cancers and the challenges that remain, it creates a shared framework for thinking about where new approaches, including AI, may have the greatest impact.
Ultimately, progress in childhood cancer will continue to come from advances in biology, from well designed clinical research, and from improving access to care.

New technologies have the potential to help accelerate that progress, but they will do so as part of a much larger and more coordinated effort. The Atlas also provides a useful lens for considering how philanthropic and strategic investments can support progress across this landscape. Opportunities may include enabling better use of data, improving access to clinical trials, and supporting research that connects discovery to patient care. Any such efforts are most effective when they are grounded in the realities of the science and aligned with the needs of patients and clinicians.

AI is becoming an important tool in cancer research, particularly in how we improve diagnosis and expand access to care. Advances in imaging analysis and disease classification are helping clinicians identify cancers earlier and more accurately, while also making specialized expertise more accessible beyond major academic centers. AI also has the potential to improve how we deliver treatment. One of the biggest barriers in cancer research is connecting patients to the right clinical trials. Computational approaches can help match patients more efficiently, broaden access to trials, and accelerate the translation of promising discoveries into patient care.

Across the field, there are already more than 100 active areas of research and clinical need where better use of data could meaningfully improve outcomes for children with cancer. What remains is the work of connecting these efforts, expanding access, and ensuring that advances in science reach every patient who needs them.

Philanthropic support can play an important role in this progress by enabling collaboration, supporting data-driven research, and helping bridge the gap between discovery and clinical application.

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*🧬🧬🧬Defining | 🧬🧬Dominant | 🧬Associated — Cancer Predisposition Syndrome tiers*

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| **Leukemia & MDS** | | | | | |
| 1 | **B-cell Acute Lymphoblastic Leukemia (B-ALL)** | Malignant proliferation of B-lymphoblasts in the bone marrow. The most common pediatric cancer overall. Encompasses multiple genomic subtypes (ETV6::RUNX1, Ph-like, hyperdiploidy, hypodiploidy) that drive risk stratification and treatment. | US: ~3,200/yr<br>Global: ~75,000/yr<br>*Most common pediatric cancer (~25% of all)* | CBC with differential; bone marrow aspirate/biopsy; flow cytometry immunophenotyping; FISH/karyotype; RNA-seq for fusion genes (ETV6::RUNX1, BCR::ABL1, Ph-like panel); SNP array for copy number alterations; MRD by PCR or flow. | ▸ **Standard of Care:** Risk-stratified COG protocols (AALL0434/AALL1231). Induction: vincristine, corticosteroid, asparaginase ± daunorubicin. Consolidation/maintenance: methotrexate backbone. Ph+ ALL: TKI (dasatinib). Ph-like: TKI or JAK inhibitor per alteration. MRD-positive: blinatumomab. Relapsed: CAR-T (tisagenlecleucel). CNS prophylaxis: intrathecal chemo ± cranial RT. |
| 2 | **T-cell Acute Lymphoblastic Leukemia (T-ALL)** | Aggressive leukemia of T-lymphoblast precursors; frequently presents with mediastinal mass, high WBC, and CNS involvement. Distinct from B-ALL biologically, often treated on same risk-adapted backbone with augmented regimens. | US: ~500/yr<br>Global: ~15,000/yr<br>*~15% of pediatric ALL* | Flow cytometry (CD3, CD7, TdT); TCR gene rearrangement; karyotype; NOTCH1/FBXW7 mutation (favorable); PTEN loss, JAK/STAT alterations; MRD by PCR. | ▸ **Standard of Care:** Augmented BFM-based protocols (AALL0434). Nelarabine added for high-risk T-ALL. CNS-directed therapy intensified. Relapsed/refractory: nelarabine, venetoclax combinations; allogeneic SCT for high-risk relapse.<br><br>▸ **Emerging:** No approved CAR-T yet for T-ALL (active trials). |
| 3 | **Infant ALL / KMT2A-rearranged ALL** | Leukemia arising in children under 12 months; nearly 80% harbor KMT2A (MLL) gene rearrangements (t(4;11) most common). Characterized by hyperleukocytosis, CNS involvement, and dismal outcomes with standard therapy. | US: ~150/yr<br>Global: ~3,000/yr<br>*<5% of ALL; age <12 months* | Flow cytometry (pro-B, CD10-negative often); FISH for KMT2A breakapart; RNA-seq for partner gene; MRD by PCR of KMT2A fusion transcript. | ▸ **Standard of Care:** INTERFANT-06/COG AALL0434 infant arm: intensive chemotherapy with high-dose cytarabine. KMT2A-r: allogeneic SCT in CR1 standard for many centers. Supportive: hyperleukocytosis management.<br><br>▸ **Emerging:** menin inhibitors (revumenib (Syndax Pharmaceuticals), ziftomenib (Kura Oncology)) targeting KMT2A-r; CAR-T trials. |
| 4 | **Down Syndrome ALL (DS-ALL) 🧬** | Children with trisomy 21 have 20-fold increased ALL risk; predominantly B-ALL with JAK2/CRLF2 alterations (Ph-like biology) and GATA1 sensitivity. Higher treatment toxicity (methotrexate, infections) requires modified protocols. | US: ~200/yr<br>Global: ~4,000/yr<br>*~20-fold elevated risk in trisomy 21* | Standard ALL workup; CRLF2 overexpression by flow/FISH; JAK1/2 mutations; JAK-STAT pathway activation profiling; DS-specific MRD monitoring. | ▸ **Standard of Care:** Modified COG protocols with reduced methotrexate, augmented leucovorin rescue. Avoid excessive anthracycline. JAK inhibitor (ruxolitinib) in AALL1231 Ph-like DS-ALL arm. Infection prophylaxis intensified. SCT generally avoided unless high-risk relapse. |
| 5 | **Acute Myeloid Leukemia (AML) — Pediatric** | Malignant expansion of myeloid progenitors; encompasses multiple WHO-defined subtypes including KMT2A-r, inv(16), t(8;21), monosomy 7, and NUP98-rearranged. Second most common pediatric leukemia; intensive induction with significant toxicity. | US: ~700/yr<br>Global: ~15,000/yr<br>*2nd most common pediatric leukemia* | Bone marrow aspirate; flow cytometry; karyotype; FISH (KMT2A, inv16, t(8;21)); FLT3-ITD/TKD, NPM1, CEBPA, IDH1/2, WT1 mutations; RNA-seq for NUP98 fusions; MRD by flow. | ▸ **Standard of Care:** COG AAML1031/AAML1831: high-dose cytarabine-based induction (Ara-C + daunorubicin + etoposide). Gemtuzumab ozogamicin for CD33+. FLT3-ITD: sorafenib/gilteritinib added. Allogeneic SCT for high-risk/refractory. Relapsed: FLAG-IDA, CPX-351.<br><br>▸ **Emerging:** Venetoclax (AbbVie) + azacitidine (Bristol Myers Squibb) investigational. |
| 6 | **Acute Promyelocytic Leukemia (APL)** | Distinct AML subtype defined by PML::RARA fusion t(15;17); pathognomonic coagulopathy (DIC) is | US: ~100/yr<br>Global: ~2,000/yr<br>*~5-10% of pediatric AML* | FISH/RT-PCR for PML::RARA (urgency); peripheral smear (faggot cells); coagulation panel (DIC); | ▸ **Standard of Care:** All-trans retinoic acid (ATRA) + arsenic trioxide (ATO) — standard for low/intermediate risk; induces remission without conventional chemo. |

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| | | life-threatening at diagnosis. Highly curable with differentiation therapy; only leukemia where chemotherapy-free remission is possible. | | bone marrow confirming APL morphology; FLT3 mutation for risk stratification. | High-risk (WBC >10,000): add anthracycline. Maintenance: ATRA ± ATO cycles ×2 years. MRD negativity by RT-PCR guides treatment duration. 5-yr OS >90%. |
| 7 | **Chronic Myeloid Leukemia (CML) — Pediatric** | Myeloproliferative neoplasm defined by BCR::ABL1 t(9;22) Philadelphia chromosome. Rare in children; biology identical to adult CML but pediatric pharmacokinetics differ. Accelerated and blast phase present as acute leukemia. | US: ~100/yr<br>Global: ~2,000/yr<br>*<5% of pediatric leukemias* | CBC (leukocytosis, basophilia, left shift); FISH/PCR for BCR::ABL1; bone marrow biopsy for phase determination; Sokal/EUTOS score for risk. | ▸ **Standard of Care:** Imatinib first-line (COG AAML0031 established dosing). Dasatinib or nilotinib for resistance/intolerance. Allogeneic SCT reserved for blast phase or TKI failure. Lifelong TKI otherwise.<br><br>▸ **Emerging:** Deep molecular response (MR4.5) goal; treatment-free remission trials ongoing. |
| 8 | **Juvenile Myelomonocytic Leukemia (JMML)** 🧬 | Aggressive myelodysplastic/myeloproliferative overlap disorder of early childhood; defined by RAS pathway mutations (PTPN11, KRAS/NRAS, NF1, CBL). Characterized by monocytosis, hepatosplenomegaly, and fetal hemoglobin elevation. | US: ~25-50/yr<br>Global: ~500/yr<br>*Rare; most common in toddlers* | CBC (monocytosis >$1\times10^9$/L, blasts <20%); HbF elevation; GM-CSF hypersensitivity colony assay; PTPN11/KRAS/NRAS/NF1/CBL mutation panel; bone marrow; cytogenetics (monosomy 7 in ~25%). | ▸ **Standard of Care:** Allogeneic SCT is only curative option; 5-yr OS ~50–60% post-SCT. Pre-SCT: azacitidine (bridges to transplant, reduces spleen). CBL mutation carriers: spontaneous remission possible — watch and wait. Relapse post-SCT: DLI, second SCT. NF1-JMML: particularly aggressive, early SCT. |
| 9 | **Myelodysplastic Syndrome — Pediatric (MDS)** | Clonal bone marrow disorders with ineffective hematopoiesis and risk of AML transformation. Pediatric MDS differs from adult: hypocellular marrow (RCC subtype), high rate of germline predisposition (GATA2, SAMD9/L, Fanconi anemia, telomeropathies). | US: ~100-200/yr<br>Global: ~2,000/yr<br>*Rare in children* | Bone marrow biopsy (cellularity, dysplasia, blast count); cytogenetics (monosomy 7 most common); germline sequencing (GATA2, SAMD9/L, Fanconi DEB test, telomere length); flow cytometry; NGS myeloid panel. | ▸ **Standard of Care:** Allogeneic SCT is standard for high-risk MDS (monosomy 7, excess blasts) and predisposition syndromes. Low-risk/hypocellular: supportive care, immunosuppression (cyclosporine + G-CSF) for selected cases. Androgen therapy for telomeropathies. GATA2 deficiency: prophylactic SCT before infections. |
| 10 | **Mixed Phenotype Acute Leukemia (MPAL)** | Rare leukemia with blasts co-expressing myeloid and lymphoid markers on same or separate populations; defined by WHO criteria. B/myeloid most common; T/myeloid less common. Often harbors complex cytogenetics or KMT2A rearrangements. | US: ~50/yr<br>Global: ~1,000/yr<br>*Rare; mixed lineage marker expression* | Flow cytometry with expanded myeloid/lymphoid panel per WHO 2022 criteria; FISH; RNA-seq; karyotype; MRD by flow or PCR. | ▸ **Standard of Care:** No consensus protocol; typically ALL-type induction (vincristine/corticosteroid/asparaginase) with AML intensification cycles. Allogeneic SCT in CR1 for most patients. Blinatumomab for B/myeloid in relapse.<br><br>▸ **Emerging:** Emerging: venetoclax (AbbVie) combinations. |
| **Lymphoma** | | | | | |
| 11 | **Hodgkin Lymphoma — Classical (cHL)** | Reed-Sternberg cell-containing lymphoma arising from germinal center B cells; four subtypes (nodular sclerosis dominant in adolescents). Mediastinal involvement in ~80%; characterized by B symptoms, constitutional features. Highly curable with risk-adapted therapy. | US: ~1,200/yr<br>Global: ~50,000/yr<br>*Most common pediatric lymphoma* | PET-CT (staging and response assessment — Deauville score); excisional lymph node biopsy; IHC (CD15, CD30, CD20 variable); BMA for stage IV; early PET response (after 2 cycles) drives escalation/de-escalation. | ▸ **Standard of Care:** COG AHOD1331: ABVE-PC (doxorubicin, bleomycin, vincristine, etoposide, prednisone, cyclophosphamide); PET-adapted — rapid responders de-escalate. Brentuximab vedotin (CD30) added to frontline high-risk. Radiation: involved-site RT (ISRT) for slow/partial responders. Relapsed: brentuximab ± bendamustine; pembrolizumab; autologous SCT. |
| 12 | **Hodgkin Lymphoma — NLPHL** | Nodular lymphocyte-predominant Hodgkin lymphoma; distinct biology from classical HL — poignantly CD20+ LP (popcorn) cells, not Reed-Sternberg cells. Indolent, localized disease in most | US: ~100/yr<br>Global: ~5,000/yr<br>*~5% of pediatric HL* | Lymph node excisional biopsy; IHC (CD20+, CD15−, CD30−, OCT-2, BOB.1); PET-CT for staging; distinguish from T-cell rich large B-cell lymphoma. | ▸ **Standard of Care:** Stage IA/IIA without risk factors: chemotherapy alone (ABVD 3 cycles) or RT; some centers observe after complete excision. Advanced/relapsed: rituximab (CD20+) effective — single agent or combined with chemotherapy. Anti- |

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| | | children; late relapses occur without transformation. | | | CD20 now frontline in many protocols. ASCT rarely needed. |
| 13 | **Diffuse Large B-Cell Lymphoma (DLBCL) — Pediatric** | Aggressive mature B-cell lymphoma; pediatric DLBCL is biologically distinct from adult — NF-κB pathway activation, IRF4 rearrangements common; germinal center B-cell (GCB) type predominant. Highly chemosensitive with short intensive therapy. | US: ~200/yr<br>Global: ~5,000/yr<br>*Largest B-cell NHL subtype in children* | Excisional biopsy; IHC (CD20, CD10, BCL6, MYC, BCL2 double-expressor); FISH (MYC, BCL2, BCL6 rearrangements — double/triple-hit); cell of origin (GCB vs. ABC); PET-CT staging. | ▸ **Standard of Care:** FAB/LMB protocol (cyclophosphamide, vincristine, prednisone, doxorubicin, methotrexate) + rituximab (Childrens NHL Consortium). Short intensive therapy (4–6 months). CR rates >85%. CNS relapse prevention: IT methotrexate. Relapsed: R-ICE, R-GDP; autologous SCT;<br><br>▸ **Emerging:** CAR-T (axicabtagene ciloleucel/Yescarta (Kite/Gilead) approved ≥18 yr, trials for pediatric). |
| 14 | **Burkitt Lymphoma** | Highly aggressive mature B-cell lymphoma defined by MYC::IGH t(8;14) translocation. Three variants: endemic (EBV-associated, jaw mass), sporadic (abdominal), immunodeficiency-associated. Fastest-dividing human tumor (Ki-67 ~100%); metabolic emergency at diagnosis. | US: ~200/yr<br>Global: ~5,000/yr<br>*Endemic in sub-Saharan Africa* | Biopsy (starry sky pattern); FISH for MYC-IG rearrangement (mandatory); LDH (tumor burden marker); LP (CNS disease in 15–30%); PET-CT; HIV/EBV testing; metabolic panel (TLS risk). | ▸ **Standard of Care:** FAB/LMB96 or CODOX-M/IVAC protocol: cyclophosphamide, doxorubicin, vincristine, methotrexate, cytarabine. Rituximab added frontline. Intensive IT chemotherapy for CNS. 5-yr OS >90% in low-risk; 70–80% advanced. TLS prophylaxis critical.<br><br>▸ **Emerging:** Relapsed: rare, poor prognosis; rasburicase/Elitek (Sanofi); clinical trials. |
| 15 | **Anaplastic Large Cell Lymphoma (ALCL) — ALK+** | T-cell lymphoma defined by NPM::ALK t(2;5) or variant ALK fusions; strong CD30 positivity. Classic systemic ALCL with B symptoms, skin lesions, and multiorgan involvement. Highly curable but early relapse common after conventional therapy. | US: ~150/yr<br>Global: ~3,000/yr<br>*Most common T-cell NHL in children* | Biopsy; IHC (CD30+, CD3 variable, EMA, ALK — cytoplasmic/nuclear pattern); FISH ALK; minimal disseminated disease (MDD) by PCR for NPM::ALK in blood/marrow for MRD. | ▸ **Standard of Care:** ALCL99/COG ANHL1232: CHOP-based induction + vinblastine maintenance. MDD-positive/refractory: crizotinib (ALK inhibitor) — COG ANHL1232 arm. Relapsed: brentuximab vedotin (CD30); crizotinib; lorlatinib (next-gen ALK inhibitor). Autologous SCT for chemo-sensitive relapse. |
| 16 | **Primary Mediastinal B-Cell Lymphoma (PMBL)** | Large B-cell lymphoma originating from thymic medullary B cells; shares biology with nodular sclerosis cHL (JAK-STAT, CD30 variable). Presents as bulky mediastinal mass with SVC syndrome in adolescent females predominantly. | US: ~50/yr<br>Global: ~1,000/yr<br>*Rare; adolescent females* | Core needle or mediastinoscopy biopsy; IHC (CD20+, CD30 weak/variable, CD15−, MUM1, MAL, CD23); PET-CT (responds slowly — EOT PET critical); distinguish from cHL and T-LBL. | ▸ **Standard of Care:** DA-EPOCH-R (dose-adjusted etoposide, prednisone, vincristine, cyclophosphamide, doxorubicin + rituximab) — avoids radiation in most responders. R-CHOP + ISRT for adolescents on some protocols. PET-negative after therapy: no RT. Pembrolizumab in relapsed/refractory (PD-L1 amplification ~75%). |
| 17 | **Lymphoblastic Lymphoma — B-cell (B-LBL)** | Lymphoma version of B-ALL (WHO: >25% marrow blasts = leukemia); presents as nodal or extranodal mass without leukemic phase. Bone, skin, and lymph node involvement common in young children. Treated identically to B-ALL with ALL-based chemotherapy. | US: ~50/yr<br>Global: ~1,000/yr<br>*B-lineage precursor lymphoma* | Biopsy; flow cytometry (TdT+, CD19+, CD10+/−); FISH for common ALL fusions; bone marrow staging essential (<25% blasts distinguishes from ALL); LP. | ▸ **Standard of Care:** ALL-based protocols (COG AALL0434 or equivalent) — identical to B-ALL treatment of same risk. No benefit to lymphoma-specific therapy. CNS prophylaxis by IT methotrexate. Prognosis equivalent to corresponding B-ALL risk group. Localized: excellent outcomes. |
| 18 | **Lymphoblastic Lymphoma — T-cell (T-LBL)** | Most common pediatric lymphoblastic lymphoma; presents as anterior mediastinal mass with SVC syndrome. TdT-positive T-lineage precursor; frequent CNS involvement; aggressive natural history requires ALL-type therapy urgently. | US: ~100/yr<br>Global: ~2,000/yr<br>*~35-40% of lymphoblastic lymphoma* | CT/MRI mediastinum + whole body; biopsy (core or open); flow cytometry (TdT+, CD3+, CD7+); TCR rearrangements; NOTCH1/FBXW7 for prognosis; BMA; LP; respiratory assessment (airway compromise risk). | ▸ **Standard of Care:** BFM-based ALL protocols (AALL0434/COG) with nelarabine intensification for high-risk. Dexamethasone superior to prednisone. IT prophylaxis essential. Response assessment after induction guides SCT decision. Relapsed: nelarabine; |

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| | | | | | ▸ **Emerging:** CAR-T trials underway. 5-yr OS ~80–85% for localized disease. |
| **19** | **Primary CNS Lymphoma — Pediatric** | Rare mature B-cell lymphoma confined to CNS (brain, spinal cord, vitreoretinal); pediatric cases often associated with immunodeficiency (HIV, post-transplant, PID). De novo primary CNS lymphoma exceedingly rare in immunocompetent children. | US: ~50/yr<br>Rare globally<br>*Very rare in children; <5% of PCNSL* | Brain MRI (enhancing lesions, periventricular); stereotactic biopsy (preferred over resection); CSF cytology + flow cytometry; vitreoretinal exam; EBV PCR in CSF; HIV/immunodeficiency workup; PET-CT to exclude systemic. | ▸ **Standard of Care:** High-dose methotrexate-based regimens (adapted from adult PCNSL protocols). Rituximab added. Reduce/eliminate immunosuppression in post-transplant cases (EBV-PTLD). Whole-brain RT avoided in young children (neurotoxicity). Intrathecal therapy. Autologous SCT consolidation in select cases.<br><br>▸ **Emerging:** EBV-PTLD: EBV-specific T cells (investigational). |
| **CNS Tumors** | | | | | |
| **20** | **Medulloblastoma** | Most common malignant pediatric brain tumor; arises in cerebellum, spreads via CSF. Four molecular subgroups (WNT, SHH, Group 3, Group 4) with distinct biology, demographics, and prognosis. WNT-activated: ~95% cure; Group 3 MYC-amplified: ~50%. | US: ~500/yr<br>Global: ~15,000/yr<br>*Most common malignant pediatric brain tumor* | MRI brain/spine (craniospinal staging critical — CSF spread in 30%); LP for cytology (10 days post-op); surgical resection + histopathology; WHO grade 4; molecular subgroup by DNA methylation array, RNA-seq; MYC/MYCN FISH; cytogenetics. | ▸ **Standard of Care:** Surgery (maximum safe resection) + craniospinal radiation (CSI, 23.4–36 Gy) + carboplatin/vincristine during RT + adjuvant chemotherapy. Infant: chemo-only (avoid CSI); high-dose chemo + ASCT. Relapsed: bevacizumab, targeted per subgroup.<br><br>▸ **Emerging:** WNT: de-intensification trials. SHH: smoothened inhibitors (vismodegib/Erivedge (Genentech/Roche)) investigational. |
| **21** | **DIPG / Diffuse Midline Glioma, H3 K27M-mutant** | Universally fatal diffuse glioma of pons (or other midline structures); defined by H3K27M histone mutation. Diagnosis made by MRI + biopsy in most centers now. Radioresistant biology; median survival 9–11 months from diagnosis despite RT. | US: ~300/yr<br>Global: ~5,000/yr<br>*Median survival <2 yrs; universally fatal* | MRI (T2 diffuse pontine expansion ≥50% of pons, poorly defined margins); biopsy now standard at most centers (liquid biopsy ctDNA in CSF emerging); H3K27M IHC/sequencing; ACVR1, PPM1D, TP53, PDGFRA mutation profiling. | ▸ **Standard of Care:** Focal RT (54 Gy in 30 fractions) — standard; temporary symptom relief in most. ACVR1-mutant: BMP pathway targeted agents. Trials through PBTC/COG essential.<br><br>▸ **Emerging:** ONC201/dordaviprone (Chimerix) (dopamine receptor antagonist) active for H3K27M+ non-DIPG and DIPG — FDA Breakthrough designation. CAR-T (GD2-directed, B7H3) early trials. |
| **22** | **Diffuse Hemispheric Glioma, H3 G34-mutant** | WHO 2021-defined high-grade glioma of cerebral hemispheres; driven by H3F3A G34R/V mutations co-occurring with TP53 and ATRX mutations. Distinct from H3K27M; occurs in older children/adolescents; INTERNOS trial network active. | US: ~50/yr<br>Global: ~500/yr<br>*Rare; teens and young adults* | MRI brain; biopsy essential (cannot diagnose by imaging alone); H3G34 IHC and sequencing; ATRX loss by IHC; p53 IHC; DNA methylation profiling; MGMT promoter methylation. | ▸ **Standard of Care:** Maximal safe surgical resection; focal RT (59.4 Gy); adjuvant temozolomide (extrapolated from adult GBM data — limited pediatric evidence). Prognosis: 2-yr OS ~30–40%.<br><br>▸ **Emerging:** G34 biology confers distinct microenvironment — immunotherapy trials ongoing. Enrollment in molecular trials strongly recommended. |
| **23** | **High-Grade Glioma — non-midline, non-H3 (incl. GBM)** | Diverse group of WHO grade 3–4 gliomas in pediatric cerebral hemispheres lacking H3 mutations; includes IDH-wildtype GBM (rare in children), hemispheric MYCN-amplified, and others. Overlaps molecularly with adult GBM but with | US: ~200/yr<br>Global: ~3,000/yr<br>*Non-H3 HGG; heterogeneous* | MRI brain with contrast; biopsy/resection; DNA methylation array for classification; IDH1/2, TERT, EGFR, MYCN, CDKN2A/B; H3 wildtype confirmation; MGMT promoter methylation status. | ▸ **Standard of Care:** Maximal safe resection; focal RT (59.4 Gy) with concurrent + adjuvant temozolomide (adult Stupp protocol applied). Bevacizumab for recurrence. Median OS 12–15 months for true pediatric GBM. |

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| | | distinct genomics. | | | ▸ **Emerging:** MYCN-amplified: Aurora A kinase inhibitor (e.g., alisertib; Takeda) trials. ONC201/dordaviprone (Chimerix) and other H3-wildtype-targeted agents in trials. |
| 24 | **Low-Grade Glioma — Pilocytic Astrocytoma and WHO grade 1–2** | Most common brain tumor in children; WHO grade 1 pilocytic astrocytoma prototype — cerebellar location, BRAF::KIAA1549 fusion in ~70%. NF1-associated optic pathway glioma distinct subset. Slow-growing; often curable with surgery alone in accessible locations. | US: ~800/yr<br>Global: ~15,000/yr<br>*Most common pediatric LGG* | MRI brain/spine; biopsy (if accessible and diagnosis uncertain); BRAF::KIAA1549 fusion (FISH/RNA-seq); BRAF V600E mutation; FGFR1/2, MYB, MYBL1; NF1 germline if optic pathway; H3-negative, IDH-wildtype in most. | ▸ **Standard of Care:** Surgery (complete resection = cure for cerebellar PA). Residual/progressive: MEK inhibitor (trametinib, selumetinib — selumetinib FDA-approved for NF1-LGG). BRAF V600E: dabrafenib/trametinib (FDA-approved 2023). Chemotherapy (carboplatin/vincristine) for NF1-LGG or when targeted agents unavailable. RT deferred as long as possible in young children. |
| 25 | **Optic Pathway Glioma (OPG)** 🧬 | LGG arising along the optic nerves, chiasm, or optic tracts; majority are pilocytic astrocytoma; ~50% in NF1 syndrome patients where natural history is more indolent. Non-NF1 chiasmatic OPG has worse prognosis and more aggressive behavior. | US: ~200/yr<br>Global: ~4,000/yr<br>*15-20% of pediatric LGG; NF1-associated* | MRI brain/orbits with fat suppression; ophthalmologic exam (visual acuity, visual fields, OCT); NF1 germline testing; biopsy often deferred given diagnosis by imaging in classic NF1; BRAF fusion if biopsy obtained. | ▸ **Standard of Care:** NF1-OPG: selumetinib (MEK inhibitor) — FDA-approved 2020 for symptomatic/progressive; carboplatin+vincristine historically. Non-NF1 OPG: chemotherapy first; RT avoided (endocrine/vascular complications). Targeted: MEK inhibitors (trametinib); BRAF V600E targeted if mutant. Preserve vision as primary goal; growth monitoring for asymptomatic NF1-OPG. |
| 26 | **Pleomorphic Xanthoastrocytoma (PXA / Anaplastic PXA)** | Rare circumscribed LGG typically in temporal lobe of young patients; WHO grade 2; BRAF V600E mutation in ~60%; superficial cortical location with cystic component. Anaplastic PXA (WHO grade 3): higher mitotic activity, worse prognosis. | US: ~50/yr<br>Global: ~500/yr<br>*Rare; BRAF-driven* | MRI (cortical-based cystic/solid, leptomeningeal enhancement possible); biopsy/resection; BRAF V600E IHC + Sanger or pyrosequencing; CDKN2A/B deletion (anaplastic feature); Ki-67; distinguish from ganglioglioma and GBM. | ▸ **Standard of Care:** Surgery (gross total resection curative for PXA grade 2 in many). BRAF V600E+: dabrafenib + trametinib (active across BRAF-mutant pediatric brain tumors). Anaplastic PXA: RT post-resection; temozolomide extrapolated from HGG protocols. RT for incompletely resected grade 2. Prognosis: 5-yr OS ~80% grade 2, ~50% anaplastic. |
| 27 | **Ependymoma — Intracranial** | Glial tumor arising from ependymal cells lining ventricles; two major intracranial groups: posterior fossa (PFA and PFB methylation subtypes) and supratentorial (ZFTA-RELA fusion or YAP1 fusion). PFA: worst prognosis; recurs locally and via CSF. | US: ~250/yr<br>Global: ~4,000/yr<br>*2nd most common malignant pediatric brain tumor* | MRI brain/spine (CSF staging); surgery; DNA methylation array (PFA vs. PFB vs. ST subtypes) — WHO 2021 mandatory classification; FISH for ZFTA::RELA in supratentorial; C11orf95 IHC; H3K27me3 loss for PFA by IHC. | ▸ **Standard of Care:** Maximal safe resection (gross total resection = best prognostic factor); focal RT (59.4 Gy) post-resection for all but youngest infants. PFA: high recurrence rate; chemotherapy role limited. Second surgery for recurrence. CpG demethylation agents. Re-irradiation at relapse. 5-yr OS ~75% ST-RELA; ~50% PFA.<br><br>▸ **Emerging:** ependymoma-specific HDAC inhibitors (e.g., panobinostat [Novartis], vorinostat [Merck]); |
| 28 | **Ependymoma — Spinal** | Ependymoma arising in spinal cord (intramedullary); predominantly NF2-associated myxopapillary ependymoma at conus/filum in adolescents, or classic WHO grade 2 cervical tumors. Distinct from intracranial; generally better prognosis with surgery. | US: ~50/yr<br>Global: ~1,000/yr<br>*Rare; mainly children <5 yrs* | MRI whole spine with contrast; surgical resection + pathology; NF2 germline testing; MYCN amplification in high-grade spinal ependymoma; distinguish myxopapillary (grade 2 per WHO 2021) from classic. | ▸ **Standard of Care:** Maximal safe surgical resection; RT for residual or high-grade disease. Myxopapillary WHO grade 2: local RT if incompletely resected. NF2-associated: surveillance after GTR. Spinal ependymoma generally chemotherapy-unresponsive. Excellent prognosis with GTR: 5-yr OS >90%. |
| 29 | **Spinal Cord Astrocytoma** | Intramedullary spinal cord tumor; predominantly pilocytic or diffuse fibrillary astrocytoma in children; rare high-grade | US: ~100/yr<br>Global: ~2,000/yr<br>*Rare; pediatric and young adult* | MRI whole spine; surgical biopsy (gross resection often not possible); molecular profiling (BRAF, | ▸ **Standard of Care:** Surgery: biopsy or limited resection to preserve function. Grade 1–2: MEK inhibitor if BRAF fusion (selumetinib/trametinib); |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| | | spinal cord gliomas occur. Presents with progressive motor/sensory deficits and scoliosis. Surgery high-risk due to eloquent location. | | H3K27M); distinguish from ependymoma by IHC (GFAP+, OLIG2+, absence of EMA dot pattern). | carboplatin + vincristine. Focal RT for progressive disease. Surveillance MRI every 3–6 months. Prognosis: LGG ~90% 5-yr OS; HGG <20%.<br><br>▸ **Emerging:** H3K27M HGG: RT + ONC201/dordaviprone (Chimerix) trial. |
| 30 | **Craniopharyngioma** | Benign but locally destructive tumor of sellar/suprasellar region arising from Rathke's pouch remnants; two types: adamantinomatous (CTNNB1 β-catenin mutations, children) and papillary (BRAF V600E, older patients). Profound hypothalamic-pituitary dysfunction regardless of treatment. | US: ~150/yr<br>Global: ~3,000/yr<br>*5-10% of pediatric brain tumors* | MRI (calcified cystic sellar/suprasellar mass); ophthalmologic exam; endocrine panel (panhypopituitarism); adamantinomatous: β-catenin mutation or IHC; papillary: BRAF V600E; distinguish from Rathke cleft cyst and pituitary adenoma. | ▸ **Standard of Care:** Surgery (limited resection to spare hypothalamus) + focal RT. Intracystic interferon-alpha or bleomycin for cystic lesions. BRAF V600E papillary: dabrafenib + trametinib (major responses). Proton beam RT preferred to spare surrounding structures. Hypothalamic obesity: behavioral/metabolic management. Endocrine replacement lifelong. 5-yr OS >95%; morbidity major challenge. |
| 31 | **Intracranial Germ Cell Tumor (iGCT)** | Midline CNS GCTs arising in pineal or suprasellar region; spectrum from pure germinoma (most common, radiosensitive, curable) to non-germinomatous GCT (NGGCT: yolk sac tumor, choriocarcinoma, embryonal carcinoma — treated with chemo + RT). Dual-peak (germinoma) marker-secreting tumors. | US: ~100/yr<br>Global: ~2,000/yr<br>*~3% of pediatric CNS tumors* | MRI brain (pineal or suprasellar, bilateral in 10%); serum + CSF AFP and β-hCG (marker-secreting NGGCT may be diagnosed without biopsy if markers elevated); biopsy for marker-negative tumors; LP for CSF cytology; spine MRI. | ▸ **Standard of Care:** Germinoma: chemotherapy (carboplatin + etoposide) + reduced-dose whole-ventricular RT → near 100% cure. NGGCT: BEP (bleomycin, etoposide, cisplatin) + whole-ventricular/CSI RT; complete resection of residual teratoma. Relapsed germinoma: high-dose chemo + ASCT. Prognosis: germinoma >95%; localized NGGCT ~70–80%. |
| 32 | **Atypical Teratoid / Rhabdoid Tumor (AT/RT)** 🧬 | Highly malignant embryonal CNS tumor; defined by SMARCB1 (INI1) or rarely SMARCA4 loss; infants and toddlers predominantly; posterior fossa and cerebellopontine angle most common. Germline SMARCB1 in ~30% (rhabdoid tumor predisposition syndrome). | US: ~50/yr<br>Global: ~500/yr<br>*Mostly infants <2 yrs; very aggressive* | MRI brain/spine; surgical resection + biopsy; IHC (INI1/SMARCB1 complete loss); SMARCB1 FISH/sequencing; germline testing; distinguish from medulloblastoma, ETMR. | ▸ **Standard of Care:** High-dose multiagent chemotherapy (carboplatin, vincristine, cyclophosphamide, etoposide, high-dose MTX); intrathecal therapy; CSI RT if >3 yr; high-dose chemo + ASCT in some protocols.<br><br>▸ **Emerging:** EZH2 inhibitor (tazemetostat/Tazverik (Ipsen, formerly Epizyme)) investigational (SWI/SNF complex loss). BET inhibitor trials. 5-yr OS ~30–45%; improved in localized/older children. |
| 33 | **Choroid Plexus Carcinoma (CPC)** 🧬🧬 | Malignant WHO grade 3 tumor of choroid plexus epithelium; high CSF protein, hydrocephalus, hemorrhage common at presentation. TP53 germline mutation (Li-Fraumeni syndrome) in ~50%. Highly aggressive; surgical resection imperative. | US: ~25/yr<br>Global: ~200/yr<br>*Very rare; TP53 germline in most* | MRI brain/spine; surgical resection + histopathology (WHO grade 3); TP53 IHC/sequencing; germline TP53 testing in all children; Ki-67 >10% distinguishes from CPP; transthyretin IHC. | ▸ **Standard of Care:** Maximal safe resection (most important factor); chemotherapy (ICE: ifosfamide, carboplatin, etoposide; cyclophosphamide, vincristine); high-dose chemo + ASCT in some protocols. CSI RT for disseminated disease. TP53 germline: avoid RT if possible (secondary malignancy risk). 5-yr OS ~40–50% with aggressive multimodal therapy. |
| 34 | **CNS Embryonal Tumors (ETMR / Pineoblastoma)** | Diverse group of WHO grade 4 embryonal CNS tumors: embryonal tumor with multilayered rosettes (ETMR, C19MC amplified), pineoblastoma (pineal region, DICER1 or TP53 alterations), and others. Aggressive; CSF dissemination common. | US: ~50/yr<br>Global: ~1,000/yr<br>*Rare; infants and young children* | MRI brain/spine (staging); biopsy; C19MC amplification by FISH (ETMR-defining); DICER1 mutation; TP53; LIN28A IHC for ETMR; LP for CSF staging. | ▸ **Standard of Care:** Maximal resection + CSI RT + chemotherapy (multiagent); high-dose chemo + ASCT used in infants to defer RT. ETMR: dismal prognosis (5-yr OS ~10–20%). Pineoblastoma: better prognosis in localized disease (5-yr OS ~50%). Rare — enrollment in PBTC or SIOPE trials essential.<br><br>▸ **Emerging:** DICER1-associated pineoblastoma: |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| | | | | | emerging targeted approach. |
| 35 | **Ganglioglioma / DNET** | Low-grade neuronal-glial tumors associated with medically intractable epilepsy; ganglioglioma (WHO grade 1–2, neurons + glial, BRAF V600E ~40%), DNET (cortical, bland, multinodular). Both are benign; surgery = cure in most. Malignant transformation rare. | US: ~200/yr<br>Global: ~3,000/yr<br>*Relatively common low-grade CNS tumor* | MRI (cortical cystic-solid, temporal lobe, ±calcification); often found during epilepsy workup; biopsy/resection; BRAF V600E IHC and molecular testing; Ki-67 for grade; EEG/video-EEG for surgical planning. | ▸ **Standard of Care:** Complete surgical resection = cure in most (seizure freedom 70–80%). BRAF V600E+: dabrafenib/trametinib for unresectable or recurrent. RT rarely needed. Chemotherapy not standard. Anaplastic ganglioglioma: treat as HGG. Long-term: seizure management, neuropsychological follow-up. |
| 36 | **Astroblastoma, MN1-altered** | Rare WHO 2021-defined CNS tumor; predominantly cerebral hemispheres in children and young adults; MN1 gene fusion or translocation (with BEND2 or CXXC5). Histologically distinctive astroblastic perivascular pseudorosettes. Variable clinical behavior but often low-grade. | <20/yr US<br>Extremely rare<br>*Very recently characterized entity* | MRI (well-circumscribed, superficial, cystic component); biopsy; MN1 rearrangement by FISH or RNA-seq (mandatory WHO 2021 criterion); DNA methylation profiling; distinguish from ependymoma and HGG. | ▸ **Standard of Care:** Maximal safe surgical resection — primary treatment; GTR may be curative. RT for residual/recurrent disease. No established chemotherapy regimen. MN1 fusions represent potential therapeutic target (transcriptional dysregulation). Prognosis: generally favorable with GTR; aggressive variants described. Enrollment in rare tumor registries critical. |
| **Solid Tumors — Embryonal & Neural** | | | | | |
| 37 | **Meningioma — Pediatric** | Usually benign (WHO grade 1) dural-based tumor; pediatric meningiomas biologically distinct from adult: higher NF2-independent alterations, higher recurrence rate, association with prior radiation, and SV40/radiation causation. Grade 2–3 more frequent than adults. | US: ~50/yr<br>Global: ~1,000/yr<br>*Rare in children; benign in adults* | MRI brain/spine (dural-based, enhancing, ± calcification); biopsy; IHC (somatostatin receptor); WHO grading (grade 1–3); NF2, AKT1, SMO, TRAF7 mutations; germline NF2 evaluation. | ▸ **Standard of Care:** Surgery (Simpson grade I resection for cure). RT for grade 2–3 or residual. Somatostatin analog (octreotide) palliative for progressive inoperable. NF2 syndrome: systemic treatment with bevacizumab (hearing preservation). Prognosis: grade 1 ~95% 5-yr OS; grade 3 ~60%.<br><br>▸ **Emerging:** SMO-mutant: vismodegib/Erivedge (Genentech/Roche); AKT1-mutant: capivasertib (AstraZeneca) — targeted agents in trials. |
| 38 | **Neuroblastoma** | Malignant tumor of neural crest-derived sympathoadrenal precursors; most common extracranial solid tumor of childhood. Spectrum from spontaneous regression (infant low-risk) to highly aggressive metastatic disease (high-risk, MYCN-amplified). Adrenal primary most common. | US: ~700/yr<br>Global: ~15,000/yr<br>*Most common extracranial solid tumor in children* | Catecholamine metabolites (urine HVA/VMA); MIBG scan ($^{312}$I- or $^{123}$I-MIBG) — primary staging and response; DOTATATE PET for MIBG-negative tumors; CT/MRI primary + mets; bone marrow bilateral BMA; MYCN FISH; ALK, PHOX2B, segmental chromosomal aberrations; INRG staging. | ▸ **Standard of Care:** Risk-stratified (INRG): Low-risk: surgery ± observation. Intermediate: surgery + moderate chemo. High-risk: induction chemo (ANBL1232: cyclophosphamide, doxorubicin, vincristine alternating with cisplatin/etoposide) → surgical resection → high-dose chemo + tandem ASCT → MIBG therapy + RT + anti-GD2 immunotherapy (dinutuximab) + isotretinoin. ALK-mutant: crizotinib. 5-yr OS high-risk ~50%. |
| 39 | **Ganglioneuroblastoma** | Intermediate neuroblastic tumor between ganglioneuroma and neuroblastoma; contains both mature ganglion cells and neuroblasts. Intermixed or nodular (composite) subtypes; nodular variant has embedded unfavorable neuroblastoma component. MIBG-avid like neuroblastoma; treated per neuroblastoma risk stratification. | US: ~100/yr<br>Global: ~2,000/yr<br>*Intermediate between neuroblastoma and ganglioneuroma* | MIBG scan (avid in most); CT/MRI for primary; bone marrow aspirate/biopsy; MYCN FISH; pathology: International Neuroblastoma Pathology Classification (INPC — stroma-rich features); urine catecholamines. | ▸ **Standard of Care:** Risk stratification per COG neuroblastoma guidelines (MYCN, INPC classification, INRG stage, age, ploidy). Intermixed usually low/intermediate risk: surgery ± limited chemo. Nodular: treated per embedded neuroblastoma component risk. Anti-GD2 immunotherapy for high-risk. Prognosis: intermixed ~95% 5-yr OS. |
| 40 | **Wilms Tumor — Unilateral (Favorable Histology)** 🧬 | Most common renal malignancy of childhood; embryonal tumor of metanephric blastema. Favorable histology (FH) lacks anaplasia; treated | US: ~500/yr<br>Global: ~15,000/yr<br>*Most common renal tumor in children* | Ultrasound + CT abdomen for staging; CT chest (pulmonary mets); no biopsy before surgery (NWTS/COG) or pre-treatment | ▸ **Standard of Care:** COG: upfront nephrectomy + chemotherapy. Stage I–II FH: actinomycin-D + vincristine (18 weeks). Stage III: + doxorubicin + abdominal RT. Stage IV: + pulmonary RT if lung mets |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| | | with nephrectomy + chemotherapy. Excellent prognosis. Associated with WT1, WTX, CTNNB1 mutations and overgrowth syndromes. | | biopsy (SIOP) per protocol; pathology: FH vs. anaplastic; LOH 1p and 16q for risk stratification; germline WT1 evaluation for predisposition syndromes. | persist. SIOP: pre-op nephroblastoma chemo → nephrectomy → adjuvant. 5-yr OS >90% overall. |
| 41 | **Wilms Tumor — Bilateral / Stage V** 🧬 | Synchronous bilateral Wilms tumor requires nephron-sparing surgery to preserve renal function; 4–6 cycles of pre-operative chemotherapy to reduce tumor burden; staged separately (AREN0534). Same histologic spectrum but renal preservation is primary goal. | US: ~100/yr<br>Global: ~2,000/yr<br>*~5% of Wilms tumors; bilateral involvement* | CT abdomen (bilateral lesions, volume assessment); renal function assessment (GFR); bilateral percutaneous biopsy to confirm WT before pre-op chemo in some protocols; WT1/WT2 germline evaluation; staging per AREN0534. | ‣ **Standard of Care:** COG AREN0534: pre-operative chemotherapy (actinomycin-D + vincristine ± doxorubicin) ×6 weeks → bilateral nephron-sparing surgery → adjuvant chemotherapy per final pathology. Anaplastic histology: intensified therapy. Long-term renal monitoring. 5-yr OS ~80–85% bilateral favorable histology. |
| 42 | **Wilms Tumor — Diffuse Anaplasia** | Anaplastic Wilms tumor — either focal or diffuse; diffuse anaplasia (DA) carries significantly worse prognosis; defined by markedly enlarged hyperchromatic nuclei and abnormal mitoses throughout tumor. Requires intensified therapy (regimen DD-4A with doxorubicin ± RT); TP53 alterations frequent. | US: ~100/yr<br>Global: ~2,000/yr<br>*~5-10% of Wilms; worst prognosis subtype* | Nephrectomy specimen pathology (DA cannot be diagnosed pre-operatively reliably); TP53 mutation; LOH 1p/16q; thorough sampling required (≥1 section per cm); distinguish focal from diffuse anaplasia. | ‣ **Standard of Care:** COG: Regimen DD-4A (actinomycin-D, vincristine, doxorubicin, cyclophosphamide, etoposide) for diffuse anaplasia. Abdominal RT for stages III–IV. Pulmonary RT for lung mets. Autologous SCT for relapse in select cases. Prognosis: stage-dependent; DA stage I/II ~85% 5-yr OS; stage IV ~20–30%. |
| 43 | **Hepatoblastoma** | Malignant embryonal liver tumor arising from hepatic progenitor cells; most common liver tumor of childhood; AFP markedly elevated (>100,000 ng/mL in most). CTNNB1 (β-catenin) mutations in >80%. Responsive to platinum-based chemotherapy. | US: ~150/yr<br>Global: ~2,000/yr<br>*Most common pediatric liver tumor* | AFP (markedly elevated, age-corrected; normal in newborns — high for age); CT/MRI liver (PRETEXT staging — extent of hepatic involvement); CT chest (pulmonary mets common); biopsy if diagnosis uncertain; pathology: epithelial vs. mixed vs. small cell undifferentiated (worst prognosis). | ‣ **Standard of Care:** SIOPEL/COG AHEP0731: neoadjuvant PLADO (cisplatin + doxorubicin) ×4 cycles → surgical resection (partial hepatectomy or liver transplant for unresectable) → adjuvant chemo. PRETEXT IV/metastatic: intensified (PLADO + vincristine + 5-FU). Liver transplant for unresectable PRETEXT IV confined to liver. 5-yr OS ~80% localized; ~60% metastatic. |
| 44 | **Retinoblastoma** 🧬🧬 | Malignant tumor of retinal progenitor cells; defined by biallelic RB1 loss. Heritable (bilateral, multifocal, germline RB1) or sporadic (unilateral). Presents with leukocoria and strabismus. Highly curable when intraocular; extraocular/metastatic disease rare but lethal. | US: ~300/yr<br>Global: ~8,000/yr<br>*Most common intraocular tumor in children* | Ophthalmology exam under anesthesia (EUA); RetCam imaging; MRI orbits and brain (optic nerve involvement, CNS spread); CT avoided (radiation in RB1 germline patients increases secondary cancer risk); LP for metastatic workup; germline RB1 testing in all. | ‣ **Standard of Care:** Intraocular: intra-arterial chemotherapy (IAC — ophthalmic artery infusion of melphalan ± topotecan ± carboplatin) — primary globe-salvage. Intravitreal melphalan for vitreous seeds. Systemic chemo: carboplatin, etoposide, vincristine for bilateral/advanced. Focal: laser/cryotherapy/brachytherapy. Enucleation: for eyes with no salvage potential. External beam RT: last resort in germline RB1. |
| 45 | **Pleuropulmonary Blastoma (PPB)** 🧬🧬 | Rare intrathoracic malignancy of early childhood; DICER1 syndrome-defining tumor; three types: Type I (cystic, <2 yr, best prognosis), Type II (cystic/solid), Type III (purely solid, worst prognosis). DICER1 germline in >70%. | <25/yr US<br>Global: ~200/yr<br>*Very rare; DICER1 germline* | CT chest (cystic or solid intrathoracic mass); surgical resection + pathology; DICER1 germline testing in patient and family; whole-body imaging for metastases; distinguish from type I: congenital lung malformation vs. PPB cyst. | ‣ **Standard of Care:** Type I: complete surgical resection alone (chemotherapy may be omitted); DICER1 surveillance. Type II–III: surgery + multiagent chemo (IRS/COG: vincristine, actinomycin-D, cyclophosphamide; IVADo or modified RMS protocols) ± RT. Type II ~71%; Type III ~53%.<br><br>‣ **Emerging:** Metastatic: intensified chemo + ASCT investigational. 5-yr OS: Type I ~91%; |
| 46 | **Pancreatoblastoma** | Extremely rare malignant pancreatic tumor of young children; resembles embryonic pancreatic tissue with | <20/yr US<br>Extremely rare<br>*Rarest pancreatic malignancy* | CT/MRI abdomen (pancreatic mass, AFP elevation); Whipple resection specimen for pathology; | ‣ **Standard of Care:** Complete surgical resection primary (distal pancreatectomy or Whipple); PRETEXT-like staging. Chemotherapy (cisplatin-based |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| | | squamoid corpuscles; AFP often elevated; β-catenin mutations; associated with Beckwith-Wiedemann syndrome. Head of pancreas most common location. | | squamoid corpuscles are pathognomonic; CTNNB1 mutation; AFP for monitoring; distinguish from solid pseudopapillary neoplasm and PNET. | PLADO regimen borrowed from hepatoblastoma given AFP elevation and shared biology; also gemcitabine). RT for locally advanced. Limited data: case series only. Prognosis: localized ~70% 5-yr OS; metastatic ~30%. |
| 47 | **Osteosarcoma** | Most common primary bone malignancy of childhood; arises from malignant osteoblastic precursors producing osteoid. Metaphyses of long bones (distal femur, proximal tibia, proximal humerus) predominate. Highly chemotherapy-responsive; pulmonary metastases most common site. | US: ~800/yr<br>Global: ~15,000/yr<br>*Most common primary malignant bone tumor* | Plain radiograph (sunburst pattern, Codman's triangle); MRI primary + whole bone for skip lesions; CT chest; bone scan or FDG-PET for metastases; surgical biopsy (core needle, tract excised at resection); no molecular markers established for diagnosis; TP53/RB1 germline if young or family history. | ‣ **Standard of Care:** MAP protocol (methotrexate, doxorubicin, cisplatin) — COG AOST0331/AOST1421. Resection at week 10 (limb salvage vs. amputation); histologic necrosis guides adjuvant. Good response (>90% necrosis): same MAP. Poor response: add ifosfamide + etoposide. No targeted therapy established. Relapsed: gemcitabine, docetaxel, cabozantinib.<br><br>‣ **Emerging:** Immunotherapy under investigation. 5-yr OS: localized ~70%; metastatic ~25%. |
| 48 | **Ewing Sarcoma** | Small round cell sarcoma of bone and soft tissue; defined by EWSR1::FLI1 or variant ETS-family fusions (t(11;22) in 85%). Bone (midshaft/diaphysis) and soft tissue primaries; aggressive with early hematogenous spread. Highly chemosensitive. | US: ~300/yr<br>Global: ~2,500/yr<br>*Rare in African populations* | Plain radiograph (permeative lytic lesion + periosteal reaction — onion-skin); MRI primary; CT chest + FDG-PET/bone scan for staging; core biopsy; FISH/RT-PCR for EWSR1-ETS fusions (mandatory); rarely: EWSR1-NFATc2 or non-EWSR1 fusion variants. | ‣ **Standard of Care:** COG AEWS1031: VIDE/VAC (vincristine, ifosfamide, doxorubicin, etoposide alternating with vincristine, actinomycin-D, cyclophosphamide). Local control: wide resection ± RT (55.8 Gy for unresectable). High-risk: interval compressed chemo. Metastatic/relapsed: high-dose chemo + ASCT; irinotecan + temozolomide. GD2-directed therapy; TK216;<br><br>‣ **Emerging:** PARP inhibitors in trials (olaparib [AstraZeneca], talazoparib/Talzenna [Pfizer]). 5-yr OS: localized ~70%; lung mets ~40%; bone/marrow mets ~20%. |
| 49 | **Chondrosarcoma — Pediatric** | Malignant cartilage-producing tumor; rare in children (vs. adults); pediatric variants include mesenchymal chondrosarcoma (HEY1::NCOA2 fusion) and clear cell chondrosarcoma; extraskeletal mesenchymal chondrosarcoma also seen. Chemotherapy-resistant conventional type; mesenchymal is chemosensitive. | US: ~50/yr<br>Global: ~1,000/yr<br>*Rare in children; mostly adolescents* | MRI primary + CT chest; plain radiograph (calcified matrix); biopsy; mesenchymal: HEY1::NCOA2 by FISH/RNA-seq; IDH1/2 mutation in conventional; grade (I–III) critical for prognosis. | ‣ **Standard of Care:** Surgery (wide resection) primary and often only treatment for conventional (chemo/RT-resistant). Mesenchymal chondrosarcoma: Ewing sarcoma-like chemotherapy (IE/VAC regimens) + resection + RT. Proton RT for skull base. Prognosis: mesenchymal — 10-yr OS ~27%; clear cell — better with wide resection.<br><br>‣ **Emerging:** IDH1/2 mutant: ivosidenib/Tibsovo (Servier)/enasidenib/Idhifa (Bristol Myers Squibb) investigational. |
| **Sarcomas & Rare Solid Tumors** | | | | | |
| 50 | **Malignant Peripheral Nerve Sheath Tumor (MPNST)** 🧬 | Aggressive sarcoma of Schwann cell origin; 50% arise in neurofibromatosis type 1 (NF1) patients from plexiform neurofibromas; NF1-associated MPNST has worse prognosis than sporadic. Second most common cancer in NF1 after optic pathway glioma. High local recurrence. | US: ~100/yr<br>Global: ~2,000/yr<br>*~50% NF1-associated* | MRI (growth in plexiform neurofibroma, loss of target sign, perilesional edema); FDG-PET (SUVmax >3.5 suspicious); biopsy; IHC (loss of H3K27me3 — highly sensitive for MPNST); NF1 germline; CDK4/CDKN2A alterations; distinguish from other | ‣ **Standard of Care:** Wide surgical resection (negative margins essential — difficult in NF1 anatomic locations). RT for margin-positive/unresectable (60 Gy). Chemotherapy: ifosfamide + doxorubicin (modest response ~25%). MEK inhibitor (selumetinib) for plexiform neurofibroma reduction (FDA-approved for NF1) — MPNST trials ongoing. CDK4/6 inhibitors for CDK4-amplified. 5-yr OS: ~35–50% localized; <15% |

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| | | | | spindle cell sarcomas. | metastatic/NF1-associated. |
| **51** | **Rhabdomyosarcoma — Alveolar (ARMS)** | High-risk RMS subtype defined by PAX3::FOXO1 or PAX7::FOXO1 fusions (t(2;13) or t(1;13)); myogenic origin with poor prognosis. Extremity and parameningeal primaries common. Fusion-positive ARMS segregates to high-risk group regardless of stage. | US: ~150/yr<br>Global: ~2,000/yr<br>*Higher-risk; PAX3/7-FOXO1 fusions* | MRI primary; CT chest; bone scan/FDG-PET; bilateral BMA; LP for parameningeal; core biopsy; FISH/RT-PCR for PAX-FOXO1 fusions (mandatory risk stratification); IHC (desmin, myogenin, MyoD1 — focal in ARMS); MRD by ctDNA emerging. | ▸ **Standard of Care:** COG ARST1431/ARST0431: VAC (vincristine, actinomycin-D, cyclophosphamide) backbone; high-risk adds vinorelbine/cyclophosphamide or irinotecan. Local control: surgery (wide margins) + RT (36–59.4 Gy). SJ906: intensified therapy. Targeted: CDK4/6 inhibitors; PLK1; anti-IGF1R.<br><br>▸ **Emerging:** Metastatic high-risk: trials with high-dose chemo, ASCT. Immunotherapy trials. 5-yr OS localized ~75%; metastatic ~20%. |
| **52** | **Rhabdomyosarcoma — Embryonal (ERMS)** | More common and favorable RMS subtype; spindle cell and botryoid variants have excellent prognosis; anaplastic ERMS worse. Genitourinary, head/neck (orbit, parameningeal), and biliary primaries characteristic. LOH 11p15 with IGF2 overexpression; RAS pathway mutations. | US: ~200/yr<br>Global: ~3,000/yr<br>*Most common RMS subtype in children* | MRI primary; CT chest; bone scan; bilateral BMA for high-risk; core biopsy; IHC (desmin, myogenin diffuse, MyoD1); fusion-negative (excludes ARMS); MYCN amplification in anaplastic; RAS mutation profiling (NRAS/KRAS prognostic). | ▸ **Standard of Care:** COG risk-stratified VAC-based therapy. Low-risk: reduced VAC. Intermediate: VAC ± vincristine-irinotecan. Local control: surgery first if possible; RT for unresectable/residual (Group III). Orbital: chemo + RT (no exenteration). Genitourinary: bladder preservation goal. Spindle cell/sclerosing RMS: ALK mutation subset — crizotinib active. 5-yr OS low-risk >90%; intermediate ~70%. |
| **53** | **Synovial Sarcoma** | Soft tissue sarcoma of uncertain differentiation (not synovial origin despite name); defined by SS18::SSX1/2/4 fusion t(X;18). Biphasic or monophasic histology; extremity juxta-articular most common; highly chemosensitive compared to other adult-type sarcomas; adolescents and young adults. | US: ~200/yr<br>Global: ~2,500/yr<br>*Peak incidence 15-35 yrs* | MRI primary (triple signal appearance — cystic, solid, calcification); core biopsy; FISH/RT-PCR for SS18-SSX fusion (diagnostic); IHC (TLE1+, SS18-SSX IHC — newer); SWI/SNF complex loss by SMARCB1 (retained); SS18-SSX1 vs SSX2 subtype. | ▸ **Standard of Care:** Wide surgical resection + adjuvant RT (50–60 Gy) for margin-positive. Ifosfamide highly active (response rate ~25–30%). Neoadjuvant: ifosfamide + doxorubicin for large tumors. SY-5609 (CDK7 inhibitor) — SS18::SSX transcriptional dependency. Afami-cel (TCR-T cell therapy targeting MAGE-A4) — significant responses in SS. 5-yr OS: localized ~70–80%; metastatic ~20–30%. |
| **54** | **Desmoplastic Small Round Cell Tumor (DSRCT)** | Rare aggressive sarcoma of abdominal/pelvic serosal surfaces; defined by EWSR1::WT1 fusion t(11;22)(p13;q12); males 4× more common; peritoneal dissemination at presentation in >90%; multimodal tumor with divergent differentiation on IHC. | <25/yr US<br>Global: ~200/yr<br>*Very rare; adolescent males* | CT abdomen/pelvis (innumerable peritoneal nodules — omental cake); EWSR1::WT1 by FISH or RT-PCR (diagnostic); IHC: WT1 (C-terminus), desmin, epithelial markers co-expression (polyphenotypic); PET-CT; laparoscopic biopsy. | ▸ **Standard of Care:** Aggressive multimodal: induction VAC/IE (vincristine, actinomycin, cyclophosphamide alternating with ifosfamide, etoposide); debulking surgery (peritoneal stripping); hyperthermic intraperitoneal chemotherapy (HIPEC) at specialized centers; whole abdominopelvic RT; consolidation autologous SCT. 5-yr OS ~15–25%. WT1 peptide vaccine;<br><br>▸ **Emerging:** NTRK inhibitors (larotrectinib/Vitrakvi [Bayer], entrectinib/Rozlytrek [Roche/Genentech]) for ETV6::NTRK3 subsets investigational. |
| **55** | **Infantile Fibrosarcoma / ETV6-NTRK3 Fusion** | Locally aggressive spindle cell sarcoma of infancy and early childhood; defined by ETV6::NTRK3 fusion t(12;15); also called congenital fibrosarcoma. Histologically mimics adult fibrosarcoma but with paradoxically excellent prognosis and unique TRK sensitivity. | US: ~50/yr<br>Global: ~500/yr<br>*Infants; ETV6-NTRK3 driven* | MRI primary; core biopsy; FISH/RNA-seq for ETV6::NTRK3 fusion (diagnostic and therapeutic target); IHC (TRK pan-tropomyosin receptor kinase); Ki-67; distinguish from other spindle cell tumors of infancy. | ▸ **Standard of Care:** Larotrectinib or entrectinib (TRK inhibitors) — dramatic response rates >90%; FDA-approved for TRK fusion-positive solid tumors. Drug-first approach avoids mutilating surgery; resection after response. Chemotherapy (vincristine, actinomycin, cyclophosphamide) historically used. Surgery: limb-sparing post-TRK inhibitor response. Prognosis: 5-yr OS >90% with modern TRK inhibitor therapy. |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| 56 | **Alveolar Soft Part Sarcoma (ASPS)** | Rare soft tissue sarcoma of uncertain histogenesis; defined by ASPSCR1::TFE3 unbalanced translocation; highly vascular, slow-growing but relentlessly metastatic. Lung metastases present at diagnosis in ~25%; responds to anti-angiogenic therapy. MiT family translocation cancer. | <25/yr US<br>Global: ~200/yr<br>*Very rare; adolescents/young adults* | MRI (highly vascular — flow voids); CT chest (lung mets); biopsy; FISH/RT-PCR for ASPSCR1::TFE3; IHC (TFE3, CAIX, HMB45 weak); FDG-PET (variable uptake); PD-L1 expression. | ▸ **Standard of Care:** Surgical resection with wide margins (primary treatment). Sunitinib or pazopanib (anti-angiogenic TKIs): response rate ~40–50% — highly active in ASPS. Conventional chemotherapy ineffective. RT palliative for metastases. Proton therapy for complex locations. 5-yr OS: localized ~80%; metastatic ~40% (slow course).<br><br>▸ **Emerging:** Atezolizumab/Tecentriq (Roche/Genentech) + bevacizumab/Avastin (Roche/Genentech): significant responses (durvalumab/Imfinzi (AstraZeneca) + olaparib/Lynparza (AstraZeneca) in trials). |
| 57 | **Clear Cell Sarcoma of Soft Tissue** | Melanocytic sarcoma of tendons and aponeuroses in young adults/adolescents; defined by EWSR1::ATF1 or EWSR1::CREB1 fusions; S100+ and HMB45+ (mimics melanoma); foot and ankle most common. Aggressive: lymph node spread common. | <25/yr US<br>Global: ~200/yr<br>*Very rare; EWSR1-ATF1 fusion* | MRI primary; FDG-PET; sentinel lymph node biopsy (lymph node mets ~50%); core biopsy; FISH/RNA-seq for EWSR1-ATF1 or EWSR1-CREB1; IHC (S100+, SOX10+, HMB45+, MelanA+, pan-TRK negative); melanoma BRAF V600E negative in most. | ▸ **Standard of Care:** Wide surgical resection (primary). Sentinel lymph node evaluation. Conventional chemotherapy has minimal activity. MET amplification in subset: crizotinib. CTLA4/PD1 immunotherapy: limited data from melanoma crossover. Isolated limb perfusion for locally advanced. Prognosis: 5-yr OS ~50%; high local recurrence and late metastasis rates. Enroll in rare tumor consortia. |
| 58 | **Epithelioid Sarcoma** | Rare soft tissue sarcoma with epithelial differentiation; two types: classic (distal extremity, young adults) and proximal/axial (aggressive, SMARCB1/INI1 loss in ~90%). SMARCB1 loss opens therapeutic avenue with EZH2 inhibition. Mimics benign conditions causing diagnostic delay. | <25/yr US<br>Global: ~200/yr<br>*Very rare; distal extremities* | MRI; core biopsy; IHC (EMA+, cytokeratin+, vimentin+, INI1/SMARCB1 loss by IHC — key diagnostic); CD34+ in ~50%; distinguish from carcinoma and melanoma; SMARCB1 deletion by FISH; molecular profiling. | ▸ **Standard of Care:** Wide surgical resection (margins critical). RT for inadequate margins. Tazemetostat (EZH2 inhibitor) — FDA-approved 2020 for metastatic/locally advanced epithelioid sarcoma (first tumor type approved for EZH2 inhibitor). Doxorubicin + ifosfamide for metastatic. Sunitinib palliative. 5-yr OS: distal ~75%; proximal ~40–50%. |
| 59 | **Undifferentiated Pleomorphic Sarcoma (UPS)** | High-grade soft tissue sarcoma without recognizable line of differentiation; diagnosis of exclusion after thorough workup. Extremity and retroperitoneum most common. Rare in children; when occurring in pediatric patients, requires germline evaluation (TP53, NF1, RB1). | US: ~50/yr<br>Global: ~500/yr<br>*Rare in children* | MRI primary; FDG-PET; CT chest; core biopsy with extensive IHC panel (exclude leiomyosarcoma, liposarcoma, MPNST, RMS); MDM2 (exclude dediff liposarcoma); TP53/NF1/RB1 germline if pediatric presentation; CGH/NGS. | ▸ **Standard of Care:** Wide surgical resection + RT (neoadjuvant or adjuvant). Doxorubicin + ifosfamide chemotherapy (modest benefit). Olaratumab discontinued. Immunotherapy (pembrolizumab) for high TMB subsets. Trabectedin palliative. Germline TP53: RT minimized. 5-yr OS: localized ~60%; metastatic <20%. |
| 60 | **Desmoid Tumor / Aggressive Fibromatosis** 🧬 | Locally invasive myofibroblastic neoplasm; not metastatic but causes significant local morbidity. CTNNB1 (β-catenin) mutations in sporadic; APC germline mutation in FAP-associated. Spontaneous regression documented (~20%); watch-and-wait now first-line for most. | US: ~100/yr<br>Global: ~2,000/yr<br>*~85% APC/CTNNB1-driven; FAP-associated* | MRI (infiltrative T1-intermediate, T2-variable, enhancement variable); biopsy (CTNNB1 mutation — T41A worst prognosis, S45F intermediate); APC germline if mesenteric or family history; distinguish from fibrosarcoma and nodular fasciitis. | ▸ **Standard of Care:** Active surveillance first-line (spontaneous regression ~20%). Nirogacestat (gamma-secretase inhibitor, Notch pathway) — FDA-approved 2023 for adult desmoid; pediatric trials underway. Surgery: reserved for life-threatening/refractory (high recurrence). RT for selected unresectable. NSAID + sulindac. Methotrexate + vinorelbine/vinblastine.<br><br>▸ **Emerging:** Sorafenib/Nexavar (Bayer) (TYROSINE KINASE INHIBITOR) — FDA Breakthrough for desmoid 2023; significant PFS benefit in randomized trial. |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| **61** | **Infantile Myofibromatosis** | Rare neoplastic proliferation of myofibroblasts in infants and young children; may be solitary (localized to skin/bone/soft tissue) or multicentric (can involve viscera — high mortality). PDGFRB mutations in ~70% of familial and sporadic cases; SRF-OMD fusions in subset. | US: ~50/yr<br>Rare globally<br>*Mainly neonates and infants* | MRI or CT primary lesions; plain radiograph (lytic bone lesions); biopsy; IHC (SMA+, desmin variable, h-caldesmon+); PDGFRB mutation sequencing; germline PDGFRB evaluation in familial; distinguish from fibrosarcoma and other myofibroblastic tumors. | ▸ **Standard of Care:** Solitary localized: observation (spontaneous regression) or curettage/resection. Multicentric without visceral: watch and wait or low-dose chemotherapy (methotrexate + vinblastine). Multicentric visceral: systemic chemo (vincristine, actinomycin, cyclophosphamide); PDGFRB-targeted: imatinib (gleevec) active in PDGFRB-mutant cases — small series. Prognosis: solitary ~100%; multicentric with visceral ~70% survival. |
| **62** | **Clear Cell Sarcoma of the Kidney (CCSK)** | Rare renal tumor distinct from Wilms; characterized by bone metastases (pathognomonic); BCOR internal tandem duplications (ITD) or BCOR-CCNB3 fusions in majority; YWHAE fusions in subset. Previously called 'bone-metastasizing renal tumor of childhood.' Intermediate prognosis. | <50/yr US<br>Global: ~300/yr<br>*~1% of renal tumors; distinct from Wilms* | CT abdomen (renal mass); bone scan (metastatic workup — bone mets); biopsy/nephrectomy; pathology: branching vascular septa, clear cells; IHC (cyclin D1+, BCOR+); BCOR ITD by PCR (diagnostic); distinguish from Wilms and RTK. | ▸ **Standard of Care:** Surgery (nephrectomy) + intensive chemotherapy: NWTSG-5/COG regimen: vincristine, actinomycin-D, doxorubicin, cyclophosphamide, etoposide (UH-3). RT: flank RT for all stages ≥II. 5-yr OS ~70–80% with current therapy; late relapses warrant 5+ yr surveillance.<br><br>▸ **Emerging:** EZH inhibitors (tazemetostat/Tazverik [Ipsen]; BCOR loss of function) investigational. |
| **63** | **Rhabdoid Tumor of the Kidney (RTK)** 🧬🧬 | Highly aggressive renal malignancy of infants; SMARCB1 (INI1) biallelic loss; germline in ~35% (rhabdoid predisposition syndrome). Worst prognosis of pediatric renal tumors; rapid progression, metastatic at diagnosis in >80%. Often concurrent brain AT/RT. | <25/yr US<br>Global: ~200/yr<br>*Highly aggressive; SMARCB1-deficient* | CT abdomen (necrotic renal mass, frequently extends beyond Gerota's fascia); MRI brain (exclude synchronous AT/RT); biopsy; IHC (INI1/SMARCB1 complete loss); SMARCB1 germline sequencing; BMA; LP for CNS. | ▸ **Standard of Care:** Nephrectomy; COG AREN0321: high-dose chemo (carboplatin, cyclophosphamide, etoposide); whole lung RT for pulmonary mets; flank RT. Germline SMARCB1: family surveillance. Palliative framing for most metastatic cases at diagnosis.<br><br>▸ **Emerging:** EZH2 inhibitor (tazemetostat/Tazverik (Ipsen)) — SWI/SNF loss rationale; pediatric rhabdoid tumor trials. 5-yr OS ~25%. |
| **64** | **Renal Cell Carcinoma — MiT Family Translocation (tRCC)** | Pediatric RCC subtype defined by TFE3 (Xp11) or TFEB (6p21) translocations; distinct from adult clear cell RCC; often present in adolescents after prior chemotherapy exposure (particularly alkylating agents/topoisomerase inhibitors). Often misdiagnosed as Wilms tumor. | US: ~50/yr<br>Global: ~500/yr<br>*TFE3/TFEB fusions; adolescents* | CT/MRI abdomen; nephrectomy specimen or biopsy; TFE3 FISH (Xp11 translocation); TFEB FISH (t(6;11)); IHC (TFE3, cathepsin K, CAIX, RCC marker); AFP elevated with specific fusions; distinguish from Wilms and adult RCC. | ▸ **Standard of Care:** Surgical nephrectomy (primary treatment). Systemic: sunitinib or pazopanib (VEGFR TKI) — modest activity. mTOR inhibitors (temsirolimus) — TFEB subset may respond. No standard adjuvant chemotherapy. Prognosis: localized post-nephrectomy ~70–80% 5-yr OS; metastatic ~30%.<br><br>▸ **Emerging:** Anti-PD1 combinations (pembrolizumab/Keytruda [Merck], nivolumab/Opdivo [Bristol Myers Squibb]) emerging. |
| **65** | **Hepatocellular Carcinoma — Pediatric (HCC)** | Malignant hepatocyte tumor; in children, distinct from adult HCC — often without cirrhosis, associated with hepatitis B (endemic areas), hereditary tyrosinemia, glycogen storage disease, biliary atresia. AFP elevated (~60%). Aggressive with worse outcomes than hepatoblastoma. | US: ~50/yr<br>Global: ~2,000/yr<br>*Higher rates in HBV-endemic regions* | CT/MRI liver (vascular enhancement pattern — arterial washout); AFP; HBV/HCV serology; metabolic workup (tyrosinemia: urine succinylacetone, FAH gene); biopsy if diagnosis uncertain; staging: PRETEXT and BCLC; CT chest for mets. | ▸ **Standard of Care:** Surgery (partial hepatectomy) primary; chemotherapy response inferior to hepatoblastoma. Sorafenib (VEGFR/RAF inhibitor) first-line systemic for unresectable/metastatic (adapted from adult data). TACE for downstaging. Liver transplantation for unresectable confined to liver (Milan criteria). HBV prevention (vaccination) = primary prevention in endemic areas. 5-yr OS ~30% overall.<br><br>▸ **Emerging:** NIAID trials: atezolizumab/Tecentriq (Roche/Genentech) + bevacizumab/Avastin |

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| | | | | | (Roche/Genentech). |
| **66** | **Fibrolamellar Carcinoma (FLC)** | Distinct HCC variant; DNAJB1::PRKACA fusion (chr 19 deletion) in >90%; occurs in adolescents and young adults without cirrhosis or hepatitis; large eosinophilic hepatocytes with lamellar fibrosis. AFP normal in most; LDH and neurotensin elevated. Surgery-first approach. | US: ~200/yr<br>Global: ~1,000/yr<br>*Typically young adults 20-40 yrs* | CT/MRI liver (large central scar, calcification in ~30%); AFP normal; neurotensin, LDH elevated; biopsy; DNAJB1::PRKACA fusion by RT-PCR/RNA-seq (diagnostic); distinguish from conventional HCC and cholangiocarcinoma. | ▸ **Standard of Care:** Surgical resection (only potentially curative modality); aggressive resection + lymphadenectomy. Liver transplant for unresectable. Sorafenib limited activity. Median OS ~3–5 yr for resected; ~9 months metastatic. FLC Registry enrollment recommended.<br><br>▸ **Emerging:** DNAJB1::PRKACA represents unique target: PRKACA inhibitors (investigational; academic/early-phase sponsors) and rapamycin/sirolimus (Pfizer/Wyeth) combinations in trials (CATCH-U study). Interferon-alpha + rituximab/Rituxan (Roche/Genentech) (FLC-specific experimental). |
| **67** | **Undifferentiated Embryonal Sarcoma of the Liver (UESL)** | Highly aggressive pediatric liver tumor of mesenchymal origin; large cystic/solid mass with internal hemorrhage; presents in school-age children; AFP normal; unique among pediatric liver tumors in having no elevation of AFP and being misdiagnosed as liver abscess by CT. | <25/yr US<br>Global: ~200/yr<br>*Rare primary hepatic malignancy* | MRI (large cystic hepatic mass with internal hemorrhage — may mimic abscess); AFP normal; biopsy; IHC (vimentin+, desmin variable); DICER1 mutation evaluation; distinguish from hepatoblastoma, hepatic RMS, and angiosarcoma. | ▸ **Standard of Care:** Complete surgical resection primary. Chemotherapy: IRS/RMS-based regimens (ifosfamide, doxorubicin, vincristine) — improving outcomes significantly. Liver transplant for unresectable. With multimodal therapy (resection + chemo): 5-yr OS ~70–80% (improved from ~30% with surgery alone). DICER1 mutations: PPB family surveillance. |
| **68** | **Biliary Rhabdomyosarcoma** | RMS arising from biliary tree (most commonly common bile duct); botryoid embryonal subtype; presents with obstructive jaundice in young children — often misdiagnosed as biliary atresia initially. Excellent prognosis when diagnosed early given embryonal histology. | <25/yr US<br>Global: ~200/yr<br>*Extremely rare; biliary tree RMS* | Ultrasound + MRI liver/biliary tree (grape-like mass in bile duct); ERCP or MRCP; biopsy (percutaneous or at surgery); IHC (desmin, myogenin); distinguish from choledochal cyst and biliary atresia; staging per IRS RMS criteria. | ▸ **Standard of Care:** Chemotherapy primary (VAC/VA) to shrink tumor; surgical resection (Roux-en-Y reconstruction); RT for residual disease. Excellent prognosis given embryonal botryoid histology: 5-yr OS ~75–90%. Avoid Whipple procedure when possible. IRS Group I–II: chemotherapy alone may suffice after resection. |
| **69** | **GIST — SDH-deficient / Pediatric-type** | Gastrointestinal stromal tumor arising in pediatric patients; virtually all are SDH-deficient (SDHA/B/C/D mutations) rather than KIT/PDGFRA-mutant (adult type). Gastric location predominant; multifocal; lymph node spread (unlike adult GIST); indolent course but incurable without surgical resection. | <25/yr US<br>Global: ~200/yr<br>*Very rare; SDH-deficient GIST in children* | CT/MRI abdomen (gastric mass); FDG-PET (typically avid); EGD + biopsy; IHC (CD117+, DOG1+, SDHA/SDHB loss by IHC — diagnostic of SDH deficiency); KIT/PDGFRA mutation negative; SDH subunit germline sequencing; CT chest (pulmonary chondroma); biochemical paraganglioma screening. | ▸ **Standard of Care:** Surgical resection (primary treatment); imatinib INEFFECTIVE (KIT/PDGFRA wildtype). Sunitinib or regorafenib for progressive disease. Hypoxia-inducible factor (HIF) pathway: temsirolimus. SDH-deficient GIST: chronic condition requiring surveillance. 5-yr OS with optimal surgery >85% for localized.<br><br>▸ **Emerging:** Succinate-driven SWI/SNF complex targeting: EZH2 inhibitors investigational (tazemetostat/Tazverik [Ipsen]). Immune checkpoint: durvalumab/Imfinzi (AstraZeneca) trial. |
| **Germ Cell, Endocrine & Other Carcinomas** | | | | | |
| **70** | **Solid Pseudopapillary Neoplasm of the Pancreas (SPN)** | Low-malignant-potential pancreatic tumor almost exclusively in adolescent females; CTNNB1 (β-catenin) mutations in >95%; well-encapsulated with hemorrhagic pseudocyst formation; slow-growing; | US: ~100/yr<br>Global: ~1,500/yr<br>*Almost exclusively young women* | CT/MRI abdomen (well-encapsulated cystic-solid pancreatic mass with hemorrhage); AFP/Ca19-9 normal; EUS-FNA for diagnosis if surgery uncertain; IHC | ▸ **Standard of Care:** Complete surgical resection (distal pancreatectomy ± splenectomy, or Whipple for head lesions) curative in >95%. Minimally invasive (laparoscopic) resection preferred. Chemotherapy: not established; anecdotal cisplatin use for |

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| | | excellent prognosis with complete resection. Rare metastases. | | (nuclear/cytoplasmic β-catenin, PR+, CD10+, vimentin+); distinguish from pancreatoblastoma and PNET. | unresectable/metastatic. Gemcitabine palliative for rare metastatic cases. 5-yr OS >95% with complete resection; even liver mets resectable for cure. |
| 71 | **Sacrococcygeal Teratoma — Malignant** | GCT of sacrococcygeal region; most common GCT in neonates and infants; spectrum from benign mature teratoma to malignant yolk sac tumor (endodermal sinus tumor); AFP markedly elevated in malignant components. Type I (external) vs. Types II–IV (presacral/internal). | <25/yr US malignant<br>Global: ~500/yr<br>*1 in 40,000 births; ~15% malignant* | Prenatal ultrasound (large external mass); AFP (markedly elevated, age-corrected); CT/MRI pelvis for surgical planning; resection specimen pathology; β-hCG; distinguish benign from malignant by AFP and histology. | ‣ **Standard of Care:** Complete resection with coccygectomy (prevents recurrence). Malignant (yolk sac): BEP (bleomycin, etoposide, cisplatin) or JEB (bleomycin, carboplatin, etoposide) post-resection. AFP monitoring for recurrence. Benign resection alone sufficient. 5-yr OS: malignant localized ~90%; metastatic ~50–60%. |
| 72 | **Ovarian Germ Cell Tumor — Malignant** | Dysgerminoma, yolk sac tumor, immature teratoma, and mixed GCT of ovary; most common ovarian malignancy in girls and adolescents. AFP elevated in yolk sac; β-hCG in choriocarcinoma; dysgerminoma: highly radiosensitive and chemosensitive with excellent prognosis. | US: ~100/yr<br>Global: ~2,000/yr<br>*Peak incidence: 10-14 yrs* | Pelvic ultrasound + CT abdomen/pelvis; AFP, β-hCG, LDH; surgical staging (unilateral salpingo-oophorectomy preferred — fertility preservation); pathology with GCT subtype; isochromosome 12p in most. | ‣ **Standard of Care:** Fertility-sparing surgery (unilateral oophorectomy) for stage I. BEP chemotherapy (stage IC+). Dysgerminoma: highly BEP-sensitive; surveillance after stage IA gross resection. Immature teratoma grade 3: adjuvant BEP. Consolidation: surgery for residual mature teratoma (growing teratoma syndrome). 5-yr OS: >95% localized; ~75% metastatic. |
| 73 | **Testicular Germ Cell Tumor — Pediatric** | GCT of testes in pediatric males; two distinct populations: prepubertal (yolk sac tumor pure or mature teratoma — isochromosome 12p negative) and postpubertal (classic adult GCT biology — mixed/seminoma, 12p+). Prepubertal type: low recurrence, AFP-monitored. | US: ~150/yr<br>Global: ~2,500/yr<br>*Bimodal: infants and teens* | Scrotal ultrasound; AFP (age-corrected — high in neonates normally); β-hCG; radical inguinal orchiectomy (scrotal approach contraindicated); staging CT abdomen/chest; isochromosome 12p (postpubertal biology); pathology. | ‣ **Standard of Care:** Stage I (prepubertal): orchiectomy alone + AFP surveillance. Stage I (postpubertal/adolescent): surveillance or carboplatin × 1–2 cycles. Stage II–IV: BEP (bleomycin, etoposide, cisplatin) × 3–4 cycles. Retroperitoneal lymph node dissection for residual masses. 5-yr OS: >95% stage I–II; ~70–80% metastatic. |
| 74 | **Mediastinal Germ Cell Tumor** | Primary mediastinal GCT (anterior mediastinum); nonseminomatous most common in pediatric; high AFP and/or β-hCG; presents with chest pain, SVC syndrome, dyspnea. Worse prognosis than gonadal GCT (IGCCCG poor-risk category); associated with Klinefelter syndrome. | US: ~50/yr<br>Global: ~500/yr<br>*Rare; mediastinum primary* | CT chest (anterior mediastinal mass); AFP, β-hCG, LDH; biopsy (anterior mediastinoscopy or CT-guided); avoid complete resection upfront; staging; Klinefelter karyotype if young male; echocardiogram (SVC syndrome). | ‣ **Standard of Care:** BEP × 4 cycles (poor-risk protocol) + surgical resection of residual mass. High-dose chemo + ASCT for refractory disease. Resect residual: mature teratoma or necrosis — growing teratoma syndrome. 5-yr OS ~50–60% (worse than gonadal GCT). Secondary leukemia surveillance in Klinefelter post-chemo. |
| 75 | **Mixed Germ Cell Tumor** | GCT with multiple histologic components (e.g., teratoma + yolk sac + embryonal carcinoma + choriocarcinoma); most common GCT in adolescent males (testicular) and females (ovarian); marker profile reflects predominant malignant component; treatment as per highest-risk component. | US: ~50/yr<br>Global: ~1,000/yr<br>*Multiple GCT elements combined* | AFP and β-hCG (both markers); orchiectomy or oophorectomy specimen; staging CT; IGCCCG risk classification (good/intermediate/poor) based on AFP level, β-hCG, LDH, presence of non-pulmonary visceral mets. | ‣ **Standard of Care:** BEP × 3 cycles (good risk) or × 4 cycles (intermediate/poor). Residual resection after BEP. High-dose chemo + ASCT for refractory. 5-yr OS: good-risk >90%; intermediate ~75%; poor-risk ~50%. Growing teratoma syndrome: resection mandatory. |
| 76 | **Papillary Thyroid Carcinoma — Pediatric** | Most common thyroid cancer in children; pediatric PTC biologically distinct from adult: more often multifocal, bilateral, lymph node-positive, and pulmonary metastatic — yet paradoxically excellent long-term prognosis. RET fusions predominate (especially post-radiation); BRAF V600E less common than adult. | US: ~500/yr<br>Global: ~8,000/yr<br>*Radiation exposure increases risk* | Thyroid ultrasound + US-guided FNA; total thyroidectomy specimen; IHC (CK19, HBME1, galectin-3); RET fusion (RT-PCR/RNA-seq); BRAF V600E; neck CT/MRI for nodes; $^{131}$I whole body scan post-thyroidectomy; Tg monitoring. | ‣ **Standard of Care:** Total thyroidectomy + central neck dissection; RAI ($^{131}$I) for metastatic/high-risk. RET fusion: selpercatinib or pralsetinib (FDA-approved for RET fusion-positive thyroid cancer). BRAF V600E: dabrafenib + trametinib. TSH suppression (levothyroxine). 5-yr OS >98%; late recurrence requires surveillance ×20+ yr. |

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| 77 | **Follicular Thyroid Carcinoma — Pediatric** | Second most common pediatric thyroid malignancy; vascular and capsular invasion define malignancy (cannot diagnose on FNA — requires excision); minimally invasive vs. widely invasive. DICER1 mutations in pediatric follicular thyroid cancer; RAS mutations in some. | US: ~100/yr<br>Global: ~1,500/yr<br>*Less common than papillary subtype* | Thyroid ultrasound + FNA (Bethesda IV–V — indeterminate or suspicious for follicular neoplasm); diagnostic hemithyroidectomy required; histopathology (capsular + vascular invasion); DICER1 sequencing; RAS mutation; distinguish from follicular adenoma. | ‣ **Standard of Care:** Hemithyroidectomy for low-risk minimally invasive; completion thyroidectomy for widely invasive or mets. RAI ($^{131}$I) for metastatic disease. Sorafenib/lenvatinib for RAI-refractory. DICER1 germline: surveillance for PPB family tumors. 5-yr OS: minimally invasive ~99%; widely invasive ~70%. |
| 78 | **Medullary Thyroid Carcinoma — Pediatric (MEN2/RET)** 🧬🧬🧬 | C-cell malignancy secreting calcitonin; virtually all pediatric MTC is RET germline mutation-associated (MEN2A or MEN2B); MEN2B presents in infancy with ropey nerves, marfanoid habitus. Prophylactic thyroidectomy timing based on RET mutation codon. | US: ~50/yr<br>Global: ~500/yr<br>*MEN2A/2B; RET germline mutations* | Calcitonin (markedly elevated); CEA; RET germline sequencing (ALL pediatric MTC); basal and stimulated calcitonin; neck ultrasound + MRI; pheochromocytoma biochemical screening (urine metanephrines) before surgery; parathyroid calcium. | ‣ **Standard of Care:** Total thyroidectomy + central lymph node dissection (prophylactic or curative). Pheo treated before thyroid surgery. Selpercatinib or pralsetinib (RET inhibitors) for metastatic/progressive MTC — FDA-approved. Vandetanib for pediatric advanced MTC (FDA-approved 2013). External beam RT palliative. 5-yr OS: localized (post-prophylactic thyroidectomy) ~100%; metastatic ~50%. |
| 79 | **Adrenocortical Carcinoma — Pediatric (ACC)** 🧬🧬 | Malignant adrenal cortex tumor; distinct from adult ACC — TP53 germline mutation in >50% of Brazilian children (R337H founder mutation) and ~25% globally; Li-Fraumeni syndrome major risk factor. Virilization and Cushing syndrome presenting features. | <25/yr US<br>Global: ~200/yr<br>*High incidence in Brazil: TP53 p.R337H* | Urine/serum steroids (DHEA-S, cortisol, testosterone, aldosterone); CT/MRI abdomen; PET-CT; surgical resection specimen; TP53 germline sequencing in all children; IGF2 overexpression; CTNNB1 mutation; ATRX; staging per ENSAT or modified staging. | ‣ **Standard of Care:** Complete surgical resection (adrenalectomy — laparoscopic only for small tumors, open for large). Mitotane (adrenolytic): adjuvant for high-risk resected; only established systemic agent. Cisplatin + etoposide + doxorubicin + mitotane for metastatic. TP53/germline: Li-Fraumeni surveillance protocol. 5-yr OS: Stage I–II ~80%; Stage III–IV <30%. |
| 80 | **Pheochromocytoma — MIBG-avid (VHL/NF1/RET-associated)** 🧬🧬 | Catecholamine-secreting tumor of adrenal medulla chromaffin cells; pediatric pheo is predominantly hereditary (germline VHL, NF1, RET, MAX) and MIBG-avid. Hypertensive crises, headache, sweating triad. Functionally different from SDH-deficient paraganglioma. | US: ~100/yr<br>Global: ~1,000/yr<br>*Often hereditary: VHL, NF1, RET, SDHx* | 24-hr urine catecholamines and metanephrines (elevated); plasma metanephrines; $^{123}$I-MIBG scan (primary imaging, avid in >90% VHL/RET/NF1); MRI adrenals (avoids radiation — better soft tissue); germline sequencing: VHL, RET, NF1, MAX, TMEM127. | ‣ **Standard of Care:** Surgical resection (cortical-sparing adrenalectomy for bilateral to preserve adrenal function). Pre-op alpha-blockade (phenoxybenzamine) 10–14 days mandatory. Then beta-blockade. MIBG therapy ($^{131}$I-MIBG — Azedra) for unresectable/metastatic. Sunitinib palliative. CVX surveillance lifelong (hereditary). 5-yr OS: benign resected >95%; malignant metastatic ~50%. |
| 81 | **Paraganglioma — DOTATATE-avid, SDH-deficient** 🧬🧬 | Extra-adrenal paraganglioma arising in head/neck, thorax, abdomen, or pelvis; SDH gene mutations (SDHB/C/D — Carney-Stratakis syndrome) cause SDH deficiency, activating pseudo-hypoxic HIF pathway; DOTATATE PET superior to MIBG for imaging; higher metastatic risk (especially SDHB). | US: ~50/yr<br>Global: ~500/yr<br>*SDHx germline mutations common* | 24-hr urine metanephrines/catecholamines; $^{168}$Ga-DOTATATE PET-CT (primary imaging — superior to MIBG for SDH-deficient, head/neck); SDHB/C/D germline sequencing; SDHB IHC (loss = SDH deficiency); MRI primary; CT chest for thoracic PGL. | ‣ **Standard of Care:** Surgical resection primary (complete excision). SDH-deficient/SDHB: lifelong surveillance for metastases and new primaries. Unresectable/metastatic: PRRT ($^{177}$Lu-DOTATATE — Lutathera) — FDA-approved for somatostatin receptor-positive tumors; pediatric data emerging. Sunitinib; temozolomide + capecitabine for SDHB metastatic. CVX surveillance for all family members with germline SDH mutations. |
| 82 | **Neuroendocrine Tumor — Appendiceal (Carcinoid)** | Most common pediatric GI tumor at surgery; low-grade NET of appendix; majority <2 cm and cured by appendectomy; larger tumors require right hemicolectomy. Carcinoid syndrome rare in pediatric. Incidental finding at appendectomy for appendicitis in >50%. | US: ~100/yr<br>Global: ~1,000/yr<br>*Most common GI carcinoid in children* | Discovered at appendectomy (incidentally or for appendicitis symptoms); pathology (chromogranin A+, synaptophysin+, Ki-67 <2% typical); DOTATATE PET only if >2 cm or mets suspected; 5-HIAA urine for functional tumors. | ‣ **Standard of Care:** Appendectomy alone for <2 cm tumors: curative ~99%. Right hemicolectomy for ≥2 cm or mesoappendiceal invasion. Octreotide (somatostatin analog) for carcinoid syndrome (rare). PRRT for metastatic (rare in children). 5-yr OS ~100% for <2 cm localized; ~60–70% for rare metastatic disease. |
| 83 | **Neuroendocrine Tumor — Pancreatic / Duodenal** 🧬 | Functioning (insulin, gastrin, VIP, glucagon-secreting) or non-functioning | <25/yr US<br>Global: ~500/yr | Functional: fasting insulin/C-peptide (insulinoma); gastrin (gastrinoma); | ‣ **Standard of Care:** Surgical resection (distal pancreatectomy or Whipple) primary treatment. |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| | | NET of pancreas or duodenum; pediatric cases often MEN1 syndrome-associated. Insulinoma most common functioning in children. Distinct biology from appendiceal NET; DOTATATE PET primary imaging; surgical resection curative. | *Very rare in children* | VIP; glucagon; 24-hr urine 5-HIAA (non-specific); $^{68}$Ga-DOTATATE PET-CT (primary staging); endoscopic ultrasound for small lesions; MEN1 germline sequencing; CT/MRI pancreas. | Octreotide/lanreotide for symptomatic/unresectable. PRRT ($^{177}$Lu-DOTATATE) for somatostatin receptor-positive unresectable. Everolimus (mTOR inhibitor) for progressive. Sunitinib. MEN1-associated: consider observation for small non-functioning; surgical timing debated. 5-yr OS: localized ~90%; metastatic ~50–60%. |
| 84 | **Nasopharyngeal Carcinoma — Pediatric / EBV-driven** | Squamous cell carcinoma of nasopharynx; nearly universally EBV-associated in pediatric patients (>95%); distinct from adult NPC; presents with neck mass, epistaxis, cranial nerve palsies, nasal obstruction. Highly radiosensitive; chemotherapy + RT curative for most localized cases. | US: ~200/yr<br>Global: ~12,000/yr<br>*Endemic in SE Asia and North Africa; EBV-driven* | MRI nasopharynx + neck (skull base involvement); PET-CT; EBV viral load (plasma EBV DNA — diagnostic and monitoring); endoscopic biopsy; IHC (WHO type III undifferentiated = most common in children); distinguish from lymphoma (nasopharyngeal lymphoma); EBER ISH. | ‣ **Standard of Care:** Concurrent chemoradiation (cisplatin + RT 70 Gy IMRT) — standard frontline. Induction chemotherapy (TPF: docetaxel, cisplatin, 5-FU) for bulky/stage III–IV. EBV plasma DNA guidance for adaptive therapy. Anti-PD1 (pembrolizumab, nivolumab) for recurrent/metastatic — significant activity. GemCis for recurrence. 5-yr OS: ~80% localized; ~50% metastatic. |
| 85 | **Salivary Gland Carcinoma — Pediatric** | Rare malignant tumors of major or minor salivary glands in children; most common histotypes: mucoepidermoid carcinoma (CRTC1/3::MAML2 fusion — favorable when present), acinic cell carcinoma, and adenoid cystic carcinoma (MYB::NFIB); parotid most common site. Distinct from adult in higher-grade potential. | US: ~50/yr<br>Global: ~500/yr<br>*Rare; mucoepidermoid most common subtype* | MRI parotid/submandibular/minor salivary glands; FNA (Bethesda salivary gland system); surgical excision for definitive diagnosis; FISH/RT-PCR CRTC1/3::MAML2 (mucoepidermoid); MYB::NFIB (adenoid cystic); HRAS mutation (Warthin-like); grade (low/intermediate/high). | ‣ **Standard of Care:** Parotidectomy with facial nerve preservation (primary). RT for high-grade, close margins, or perineural invasion. Adenoid cystic: adjuvant RT standard; NOTCH1-targeted therapy (AI2O2 inhibitors). Mucoepidermoid MAML2-positive: excellent prognosis post-resection. Metastatic: cisplatin + 5-FU; targeted (FGFR, TRK per fusion). 5-yr OS: low-grade ~90%; high-grade ~50%. |
| 86 | **Olfactory Neuroblastoma (Esthesioneuroblastoma)** | Malignant neuroectodermal tumor of olfactory epithelium in cribriform plate; presents with anosmia, epistaxis, and nasal obstruction. Bimodal age distribution (11–20 yr and 50–60 yr). Craniofacial resection + RT standard. Slow-growing but capable of late recurrence and CSF spread. | US: ~50/yr<br>Global: ~500/yr<br>*Rare; olfactory epithelium origin* | MRI cribriform plate/anterior skull base (extends intracranially in 50%); CT skull base; PET-CT; endoscopic biopsy; IHC (synaptophysin+, chromogranin A+, S100+ sustentacular cells, cytokeratin-); Hyams grading (I–IV); distinguish from SNUC and SNEC. | ‣ **Standard of Care:** Combined craniofacial resection (endoscopic ± open) + RT (60–65 Gy IMRT). Chemotherapy: cisplatin + etoposide neoadjuvant for high-grade/advanced Kadish C–D. Temozolomide palliative for recurrence. Octreotide/PRRT if somatostatin receptor positive. Kadish C ~65%; Kadish D ~40%.<br><br>‣ **Emerging:** Anti-GD2 (dinutuximab/Unituxin (United Therapeutics)) investigational. 5-yr OS: Kadish A/B ~80%; |
| 87 | **Thymic Tumors — Pediatric (Thymoma / Thymic Carcinoma)** | Epithelial tumors of thymus in anterior mediastinum; thymoma rare in children (unlike young adults); thymic carcinoma more aggressive. Association with myasthenia gravis and other paraneoplastic syndromes in thymoma. GTF2I mutation in type A/AB thymoma. | <25/yr US<br>Global: ~300/yr<br>*Rare in children; MG-associated* | CT chest (anterior mediastinal mass with fatty infiltration); MRI; anti-AChR antibodies (myasthenia gravis screening); biopsy (CT-guided or mediastinoscopy); WHO classification (A, AB, B1/B2/B3, TC); GTF2I mutation; PD-L1 IHC; FISH for 6q deletion. | ‣ **Standard of Care:** Complete surgical resection (thymectomy) curative for encapsulated thymoma. Extended resection for invasive. RT for incompletely resected B2/B3 or thymic carcinoma. Cisplatin-based chemotherapy (CAP: cyclophosphamide, doxorubicin, cisplatin) for unresectable. Pembrolizumab + lenvatinib active in thymic carcinoma. Somatostatin analog palliative. Myasthenia gravis: pyridostigmine, IVIg, PLEX. |
| **Histiocytic, NUT, Fusion & Predisposition** | | | | | |
| 88 | **Melanoma — Pediatric** | Malignant tumor of melanocytes; in prepubertal children often arises from giant congenital nevi or as spitzoid neoplasms; postpubertal melanoma | US: ~300/yr<br>Global: ~5,000/yr<br>*Rare in young children; increasing in adolescents* | Dermoscopy; wide excision with sentinel lymph node biopsy (staging); pathology (Breslow thickness, ulceration, mitotic rate, | ‣ **Standard of Care:** Wide local excision (margins per Breslow thickness). Sentinel LN biopsy. Stage III: adjuvant pembrolizumab or nivolumab (1 yr). BRAF V600E: adjuvant dabrafenib + trametinib (FDA- |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| | | biologically similar to adult (BRAF V600E ~40–50%). Diagnosis challenging — Spitz nevus and melanoma overlap pathologically. | | regression); BRAF V600E/K; NRAS; NF1; KIT; LDH; PET-CT for staging; distinguish from atypical Spitz tumor. | approved >12 yr post pediatric trial). Stage IV: pembrolizumab, nivolumab, or dabrafenib+trametinib. IPI + NIVO combination. 5-yr OS: Stage I >95%; Stage IV ~45–50% with modern immunotherapy. |
| 89 | **Spitz Melanoma / Spitzoid Neoplasm** | Spectrum of Spitzoid melanocytic tumors ranging from benign Spitz nevus to atypical Spitz tumor (AST) to frank Spitz melanoma; kinase fusions (ROS1, ALK, NTRK1/3, MET, RET, BRAF-atypical) characterize most; distinct from conventional melanoma; responds to TRK/ALK inhibitors when fusion-driven. | US: ~200/yr<br>Global: ~2,000/yr<br>*Distinct from conventional melanoma* | Dermatopathology (architectural disorder, large epithelioid/spindle cells, Kamino bodies); FISH (6q23 loss, 11q23 gain, 8q24 gain, 9p21 loss); RNA-seq for kinase fusions (ROS1, ALK, NTRK1/3 — therapeutic targets); distinguish from conventional melanoma (BRAF V600E negative in most Spitz); comparative genomic hybridization. | ▸ **Standard of Care:** Wide local excision + sentinel LN biopsy for AST/Spitz melanoma. NTRK fusion: larotrectinib — dramatic responses reported. ALK fusion: crizotinib. ROS1: crizotinib/entrectinib. BRAF atypical fusion: dabrafenib. Avoid radical treatment for isolated SLNB-positive AST (regional mets exceedingly rare). 5-yr OS: Spitz melanoma ~75–85% even with LN involvement. |
| 90 | **Langerhans Cell Histiocytosis — Single-System (SS-LCH)** | Clonal neoplastic proliferation of Langerhans cell-lineage myeloid progenitors; single-system means one organ or system involved (bone most common — often solitary lesion, skull/vertebra); BRAF V600E in ~50–60%; skin LCH in infants often self-limited. | US: ~500/yr<br>Global: ~5,000/yr<br>*Single system/single lesion; often curable* | X-ray + MRI (lytic bone lesion, vertebra plana); whole-body MRI or FDG-PET for assessment; biopsy (CD1a+, CD207/langerin+, S100+); BRAF V600E (mutation ~50–60%); BRAF wildtype: MAP2K1, other RAS pathway mutations; distinguish from Ewing sarcoma (bone) and seborrheic dermatitis (skin). | ▸ **Standard of Care:** Solitary bone: curettage ± intralesional steroid injection (methylprednisolone); low-dose RT (6–10 Gy) for CNS-risk lesions (temporal, orbital, mastoid, sphenoid). Skin-only infant: emollient, topical steroids; 50% spontaneous remission. BRAF V600E+: vemurafenib/dabrafenib for refractory. 5-yr OS SS-LCH >99%. |
| 91 | **Langerhans Cell Histiocytosis — Multi-System (MS-LCH)** | LCH involving ≥2 organ systems; risk-organ involvement (liver, spleen, bone marrow — 'risk-positive') confers high mortality and risk of LCH-associated neurodegeneration (LCH-ND). Risk-organ negative MS-LCH: better prognosis. BRAF V600E ~55%; MAP kinase pathway universal. | US: ~200/yr<br>Global: ~2,000/yr<br>*Multi-organ involvement; higher morbidity* | FDG-PET or $^{68}$Ga-DOTATATE PET + whole-body MRI for staging; bone marrow biopsy; liver biopsy (if hepatomegaly); CSF and brain MRI (LCH-ND screening); BRAF V600E circulating cell-free DNA (cfDNA) — response monitoring and disease activity; multi-system organ function assessment. | ▸ **Standard of Care:** LCH-IV protocol: vinblastine + prednisone (6-week induction) × 2 courses; maintenance vinblastine/prednisone ×12 months for risk-positive. BRAF V600E risk-refractory/RO-positive: vemurafenib, dabrafenib, trametinib — dramatic responses. Clofarabine or cladribine for refractory. SCT for refractory risk-positive.<br><br>▸ **Emerging:** LCH-ND: cytarabine + cladribine; clinical trials. |
| 92 | **NUT Carcinoma** | Aggressive undifferentiated carcinoma defined by NUTM1 gene rearrangement (usually BRD4::NUTM1 t(15;19)); primarily midline location (mediastinum, head/neck, lung); rapidly progressive; often misdiagnosed as anaplastic carcinoma or lymphoma. BET inhibitor-sensitive. | ~50/yr US all ages<br>Global: ~500/yr<br>*NUT gene fusion; aggressive* | Biopsy; NUT IHC (>50% nuclear speckled staining — highly sensitive/specific); FISH or RT-PCR for BRD4::NUTM1 (or BRD3, NSD3 partners); distinguish from Ewing sarcoma, DLBCL, and anaplastic carcinoma by NUT IHC; NUTM1 family partners by RNA-seq. | ▸ **Standard of Care:** Platinum-based chemotherapy + RT (modest response). Intensive multiagent chemo (carbo/taxol; ICE regimens). Surgical resection where feasible. HDAC inhibitor combinations. NUT Carcinoma Registry: enrollment essential given rarity. 5-yr OS ~15–20% overall.<br><br>▸ **Emerging:** BET bromodomain inhibitor (BETi) trials: OTX015/birabresib (Merck KGaA) and other BET inhibitors — early phase. |
| 93 | **SMARCB1-Deficient Tumors — Extracranial Malignant Rhabdoid Tumor (eMRT)** 🧬🧬 | Highly aggressive extracranial rhabdoid tumors (kidney, liver, soft tissue) with biallelic SMARCB1 inactivation; germline in ~30–40% (rhabdoid predisposition syndrome with concurrent AT/RT risk). Infants and toddlers. EZH2 inhibition is | <25/yr US<br>Global: ~200/yr<br>*Extracranial rhabdoid tumors; SMARCB1-deficient* | MRI primary site; brain/spine MRI (exclude concurrent AT/RT); BMA; biopsy; IHC (INI1/SMARCB1 complete loss); SMARCB1 sequencing (somatic biallelic); germline SMARCB1; distinguish | ▸ **Standard of Care:** Multiagent chemo (carboplatin, cyclophosphamide, etoposide, doxorubicin); high-dose consolidation; focal RT. BET inhibitors.<br><br>▸ **Emerging:** Tazemetostat/Tazverik (Ipsen) (EZH2 |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| | | mechanistically rational therapy. | | from other small round blue cell tumors. | inhibitor) — SWI/SNF complex reconstitution rationale; pediatric rhabdoid trials. Immunotherapy trials (PD-L1 expressed in subset). 5-yr OS <25% metastatic; localized ~50%. |
| 94 | **CIC-Rearranged Sarcoma** | Small round cell sarcoma defined by CIC::DUX4 (or CIC::FOXO4) fusions; now WHO 2020-classified separately from Ewing sarcoma; more aggressive than Ewing; soft tissue predominance; WT1 and ETV4 expression by IHC; prognosis worse than Ewing despite similar histology. | US: ~50/yr<br>Global: ~300/yr<br>*Recently characterized; CIC::DUX4 fusion* | Biopsy; FISH for CIC rearrangement; RNA-seq for CIC::DUX4 fusion (preferred — more sensitive); IHC (WT1+, ETV4+, CD99 weak, EWSR1 fusion-negative by FISH); BCOR-CCNB3 and EWSR1 fusions must be excluded. | ‣ **Standard of Care:** Ewing-like chemotherapy (VAC/IE or similar); local control with surgery + RT. Standard chemotherapy less effective than in classic Ewing. Clinical trial enrollment via COG/SARC essential. 5-yr OS ~40–50%; worse than Ewing Sarcoma.<br><br>‣ **Emerging:** CDK12 inhibition, YAP pathway targeting (CIC is YAP/TAZ repressor — loss activates oncogenes) investigational. |
| 95 | **BCOR-Altered Tumors (BCOR-CCNB3 Sarcoma / BCOR ITD)** | Small round cell sarcoma defined by BCOR-CCNB3 fusion (X-chromosome inversion, predominantly young males) or BCOR internal tandem duplication (ITD — also defines most CCSK); Ewing-like but distinct; BCOR-CCNB3: bone and soft tissue in adolescent males; epigenetic target (PRC1.1 complex). | <25/yr US<br>Global: ~200/yr<br>*BCOR-CCNB3 or BCOR ITD driven* | Biopsy; BCOR-CCNB3 fusion by RT-PCR or RNA-seq; FISH for X chromosome inversion (BCOR-CCNB3); IHC (BCOR overexpression+, cyclin B3+, CD99 variable); EWSR1 fusion negative; bone marrow and staging CT. | ‣ **Standard of Care:** Ewing-like chemotherapy protocols (VAC/IE) — used empirically; some evidence of response. Surgery + RT for local control. Prognosis: localized ~70–80% 5-yr OS; metastatic ~40%.<br><br>‣ **Emerging:** EZH2 inhibitor (tazemetostat/Tazverik [Ipsen]); BCOR is PRC1.1 component — related to SWI/SNF complex) — rationale for epigenetic targeting; trials underway. Clinical trial enrollment recommended. |
| 96 | **ALK+ Inflammatory Myofibroblastic Tumor (IMT)** | Myofibroblastic neoplasm with inflammatory background; ALK rearrangements in ~50% (ALK::TPM3, ALK::TPM4, etc.); epithelioid inflammatory IMT variant worse prognosis; pulmonary, mesenteric, retroperitoneal sites in children. Dramatic response to crizotinib. Rare distant metastasis (unlike IMT-like tumors). | US: ~100/yr<br>Global: ~1,000/yr<br>*Can occur at any site; ALK-driven* | CT primary; core biopsy; IHC (ALK — granular cytoplasmic); ALK FISH for rearrangement; RNA-seq for fusion partner; distinguish from RMS, DSRCT, and EBV-associated SMT in immunocompromised; plasma ALK ctDNA emerging. | ‣ **Standard of Care:** Complete surgical resection primary (curative if resectable). ALK+: crizotinib — dramatic responses >50%; alectinib for crizotinib-resistant. NSAIDs (COX-2 inhibition) — anti-inflammatory palliation. ALK-negative: surgery ± chemotherapy. Epithelioid ALK+ variant: more aggressive — systemic ALK inhibitor + surgery. 5-yr OS: localized ~90%; metastatic ~60%. |
| 97 | **NTRK Fusion Tumors (TRK Fusion Cancer)** | Pan-tumor category defined by NTRK1/2/3 gene fusions driving constitutively active TRK kinase signaling; multiple histotypes in pediatrics (infantile fibrosarcoma, gliomas, papillary thyroid, secretory breast carcinoma, MASC); larotrectinib and entrectinib FDA-approved agnostically. | US: ~50/yr<br>Global: ~500/yr<br>*Pan-tumor therapy target; larotrectinib approved* | Comprehensive tumor profiling (RNA-seq or hybrid capture DNA sequencing for fusions); IHC (pan-TRK — screening, not confirmatory); FISH for specific NTRK rearrangements; tumor-agnostic indication requires NTRK fusion documentation; resistance mutations (NTRK kinase domain) at progression. | ‣ **Standard of Care:** Larotrectinib (Vitrakvi) or entrectinib (Rozlytrek) — FDA-approved tumor-agnostic for TRK fusion-positive solid tumors; overall response rate ~75% pediatric/adult. Larotrectinib: pediatric-specific dosing approved. Next-generation TRK inhibitors (selitrectinib, repotrectinib) for resistance. Duration: continue until progression. 5-yr OS: infantile fibrosarcoma with TRK inhibitor >90%. |
| 98 | **DICER1 Syndrome Tumors** 🧬🧬🧬 | Germline DICER1 mutation causes a cancer predisposition syndrome; associated tumors: pleuropulmonary blastoma, cystic nephroma/Wilms, ovarian sex cord-stromal tumors (Sertoli-Leydig cell tumor — SLCT), cervical embryonal RMS, pineal pineoblastoma, | US: ~50/yr<br>Global: ~500/yr<br>*Germline DICER1; multiple tumor types* | Germline DICER1 sequencing (all PPB patients, SLCT, cervical RMS, pineoblastoma, cystic nephroma); somatic hotspot mutations (E1705, E1709) in tumors; surveillance protocol: annual chest CT age 0–8 yr (PPB), pelvic ultrasound (SLCT), | ‣ **Standard of Care:** Tumor-specific treatment per histotype (PPB per type; SLCT: surgery; thyroid carcinoma: thyroidectomy + RAI; RMS: VAC protocol). No DICER1-specific targeted therapy approved yet. Surveillance protocol per DICER1 Consortium guidelines. Genetic counseling for all family members. International DICER1 Registry enrollment for all |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
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| | | multinodular goiter/thyroid carcinoma, pituitary blastoma. Somatic DICER1 'hotspot' second hits drive tumorigenesis. | | thyroid ultrasound. | affected families. |
| 99 | **CMMRD-Associated Tumors** 🧬🧬🧬 | Constitutional mismatch repair deficiency syndrome (biallelic MMR gene mutations: MLH1, MSH2, MSH6, PMS2); causes childhood-onset brain tumors (high-grade glioma), colorectal cancer, lymphoma, and other cancers; ultra-hypermutation confers exceptional immunotherapy sensitivity. | <25/yr US<br>Extremely rare<br>*Constitutional mismatch repair deficiency* | IHC (MMR proteins — all four lost in CMMRD tumor); tumor mutational burden (ultra-high: >100 mut/Mb in most); MSI-H; germline MMR gene sequencing (biallelic); Lynch syndrome exclusion (monoallelic); sibling germline testing; café au lait macules >5. | ‣ **Standard of Care:** PD-1 blockade (nivolumab, pembrolizumab) — dramatic responses in ultra-hypermutated CMMRD tumors regardless of histotype; FDA-approved for MSI-H/dMMR tumors (tumor-agnostic). Ipilimumab + nivolumab for refractory. Histotype-specific chemotherapy as bridge. Surveillance colonoscopy/MRI protocol for CMMRD families. Germline counseling for siblings. |
| 100 | **Chordoma (Pediatric/Infantile)** 🧬🧬 | Malignant tumor of notochordal remnants; clivus predilection in children vs. sacrum in adults. Infantile/pediatric chordoma frequently SMARCB1-deficient (INI1-loss) unlike adult counterpart; locally destructive, prone to local recurrence. Poorly differentiated (SMARCB1-loss) variant in infants — extremely aggressive and distinct from conventional chordoma. | US: ~50/yr<br>Global: ~300/yr<br>*SMARCB1-deficient infantile variant: <10/yr globally* | MRI skull base/spine (T2 hyperintense, destructive midline mass); core biopsy; IHC (brachyury T — pathognomonic, SMARCB1/INI1 loss in infantile variant); SMARCB1 FISH; distinguish from chondrosarcoma (brachyury negative) and AT/RT (clinical context); RNA-seq for rare TBXT fusions. | ‣ **Standard of Care:** En bloc resection with negative margins — curative if achieved; proton beam radiotherapy (60-70 CGE) preferred over photon RT; skull base requires proton or carbon ion therapy. Infantile SMARCB1-deficient: high-dose multiagent chemotherapy (carboplatin, cyclophosphamide, etoposide) + proton RT. ‣ **Emerging:** EZH2 inhibitor (tazemetostat) for SMARCB1-deficient pediatric chordoma (rhabdoid predisposition overlap); CDK4/6 inhibitors (palbociclib) in adult trials; EGFR inhibitors (afatinib); PD-L1 checkpoint blockade; RARE-Peds molecular matching trials. |
| 101 | **Erdheim-Chester Disease (ECD)** 🧬🧬 | Non-Langerhans histiocytosis defined by foamy CD68+/CD1a- histiocytes with bilateral long bone sclerosis and systemic infiltration; BRAF V600E in ~60%; pediatric cases documented but rare; systemic manifestations include retroperitoneal fibrosis ('hairy kidney'), CNS xanthogranulomas, and cardiovascular sheathing. Biologically and clinically distinct from LCH. | US: <50/yr all ages<br>Global: ~500/yr all ages<br>*Pediatric (<18 yr): <10/yr; male predominance* | PET-CT (FDG-avid bilateral diaphyseal/metaphyseal long bone cortical sclerosis — pathognomonic pattern); biopsy; IHC (CD68+, CD163+, CD1a-, Langerin-); BRAF V600E molecular confirmation; NGS for MAP2K1, PIK3CA, NRAS, ARAF; ECHO for cardiac sheathing; MRI brain for CNS involvement. | ‣ **Standard of Care:** BRAF V600E+: vemurafenib (FDA-approved 2017) — first-line standard; dabrafenib + trametinib for intolerance or progression. BRAF WT: PEG-interferon-alpha-2a first-line; anakinra (IL-1R antagonist) for refractory. CNS disease: intensify systemic therapy; steroids for acute neurological compromise. ‣ **Emerging:** MEK inhibitors (cobimetinib, trametinib) for BRAF WT MAP2K1-mutant ECD; cladribine/clofarabine for chemotherapy-eligible refractory; vemurafenib + cobimetinib combination; NCI ECD registry and Histio Coalition enrollment recommended for all pediatric cases. |
| 102 | **Adamantinoma** | Rare low-grade malignant biphasic bone tumor; >90% arise in anterior tibial cortex; affects adolescents and young adults; composed of epithelial nests (CK5/6+, EMA+) within osteofibrous dysplasia-like stroma. Osteofibrous dysplasia considered benign precursor. Late metastases (lymph nodes, lungs, bone) in ~20% — may occur decades after diagnosis; dedifferentiated variant rare but aggressive. | US: ~50/yr<br>Global: ~200/yr<br>*<1% of primary bone tumors; peak age 10-30 yr* | Plain radiograph (anterior tibial 'soap bubble' cortical lytic lesion — pathognomonic location/morphology); MRI for medullary and soft tissue extension; core or incisional biopsy; IHC (CK5/6, CK14, EMA+; brachyury negative); EWSR1 FISH (negative — excludes Ewing); molecular profiling to exclude NTRK fusion sarcomas. | ‣ **Standard of Care:** Wide local resection with negative margins — segmental tibial resection with fibula autograft, cortical allograft, or endoprosthetic reconstruction; limb salvage preferred over amputation. No established chemotherapy or radiation role in classic adamantinoma. Surveillance: chest CT annually x10 yr for late pulmonary metastases. 5-yr OS: ~90% localized; ~50% metastatic. ‣ **Emerging:** Dedifferentiated adamantinoma: Ewing-like chemotherapy (VDC/IE); CDK4/6 inhibitors (frequent CDK4 amplification); NTRK inhibitors if fusion present. FUS-TFEB fusion subset: RET inhibitors investigational. Metastatic: molecular tumor board referral and clinical trial enrollment. |
| 103 | **RMS — Spindle Cell/Sclerosing** | Distinct WHO 2020 RMS subtype. Infant spindle cell RMS: paratesticular predominance, SRF::NCOA2 or VGLL2 | US: ~50/yr<br>Global: ~300/yr<br>*~5% of all RMS; infant vs.* | MRI primary + staging MRI brain/spine (parameningeal); PET-CT; bone marrow biopsy; IHC | ‣ **Standard of Care:** Infant spindle cell (SRF/VGLL2+): COG low-risk protocols (VA or VAC) — excellent outcomes, minimize toxicity. MYOD1-mutant: |

| # | Condition | Description | Frequency (US / Global) | Diagnosis | Treatment |
|---|---|---|---|---|---|
| | | fusions, excellent prognosis. Older child/adult MYOD1 L122R-mutant spindle/sclerosing RMS: dismal prognosis (>50% mortality), poor chemotherapy response, head-neck and parameningeal sites. Overlap with embryonal RMS on routine IHC requires molecular confirmation. | *MYOD1-mutant = opposite prognoses* | (desmin+, MyoD1, myogenin — scattered/weak vs. diffuse in ARMS); FOXO1 FISH (negative — excludes ARMS); MYOD1 L122R hotspot sequencing (required for diagnosis and risk stratification); SRF/VGLL2/NCOA2 FISH or RNA-seq for infant spindle cell subtype. | COG ARST1431 intensified VAC/VI; early surgical resection if feasible; RT per standard site-based dosing. Parameningeal: intrathecal chemotherapy consideration per COG protocol. ▸ **Emerging:** MYOD1-mutant: PI3K/AKT/mTOR inhibitors (frequent co-mutated PIK3CA); CDK4/6 inhibitors; immunotherapy (pembrolizumab trials); targeted therapy per co-occurring alterations on NGS. COG ARST2032 enrollment strongly recommended for all MYOD1-mutant cases given poor outcomes on standard therapy. |
| **104** | **Rosai-Dorfman Disease (RDD)** 🧬 | Non-Langerhans histiocytosis defined by massive painless lymphadenopathy and emperipolesis (lymphocytes engulfed within histiocytes) — pathognomonic on H&E. Pediatric cases ~20% of all RDD. Sporadic (~70%); SLC29A3 germline in Faisalabad histiocytosis/H syndrome; MAP kinase mutations (KRAS, MAP2K1, BRAF) in ~30% of sporadic cases. Usually self-limiting; extranodal CNS, skin, and bone RDD may require therapy. | US: ~100/yr all ages<br>Global: ~500/yr all ages<br>*Pediatric: ~20/yr; predominantly young adults; males > females* | Excisional lymph node or lesional biopsy; IHC (S100+, CD68+, CD163+, CD1a-, Langerin- — distinguishes from LCH); emperipolesis on H&E pathognomonic; SLC29A3 germline sequencing in pediatric cases; somatic KRAS/BRAF V600E/MAP2K1 sequencing; PET-FDG for extranodal extent; MRI brain/spine for CNS involvement; distinguish from lymphoma, LCH, and ECD. | ▸ **Standard of Care:** Nodal-only or self-limited: observation — spontaneous regression in majority. Symptomatic/progressive nodal: low-dose prednisone. Isolated extranodal (compressive): surgical resection or corticosteroids. CNS RDD: dexamethasone; surgical decompression for mass effect; whole brain RT as last resort. ▸ **Emerging:** MEK inhibitors (cobimetinib, trametinib) for MAP2K1/KRAS-mutant refractory RDD — case series responses; BRAF V600E+: vemurafenib/dabrafenib-trametinib; cladribine/clofarabine for chemotherapy-eligible refractory. SLC29A3-associated: HSCT case reports. Histio Coalition registry enrollment recommended. |

sources: COG, SIOPE, PBTC, WHO 2021/2022 Classification, INRG, SEER, published literature. Incidence figures are U.S. annual estimates. Treatment reflects current standard of care; enrollment in clinical trials recommended for all rare and high-risk pediatric cancers.